Hyper-Local Aviation Weather Forecasting and Verification System for Sri Lankan Monsoon Conditions

INDIVIDUAL THESIS

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ABSTRACT

This individual thesis documents the QNH, dewpoint and relative humidity (RH) prediction component of a hyper-local aviation weather forecasting and verification system for Bandaranaike International Airport (VCBI), Sri Lanka. The component addresses the need for short-term, airport-specific meteorological information under the country's Southwest Monsoon, Northeast Monsoon and inter-monsoon conditions. Historical VCBI observations covering 2019–2026, reported as more than 120,000 samples for the overall system, are chronologically ordered and processed to create time-based and rolling features. Separate LightGBM models forecast QNH and dewpoint three hours ahead. RH is derived from the predicted dewpoint and predicted dry-bulb temperature using the August–Roche–Magnus relationship rather than a separate RH machine-learning model. The wider system obtains METAR observations, stores observations and forecasts, and compares time-stamped predictions with later observations for verification. The supplied research paper reports a dewpoint coefficient of determination (R^2) of approximately 0.66 and mean absolute error (MAE) of approximately 0.58 °C. The reported QNH MAE is approximately 1.8 hPa. A separate RH error score is not provided in the paper and is therefore not claimed here. The individual contribution covers dataset preprocessing, LightGBM model training, RH formula implementation, model evaluation and backend integration. The results indicate that the module can contribute useful forecast inputs to airport-level situational awareness; however, operational use requires further season-specific validation, transparent data-quality handling and documented verification of RH outputs.

Keywords: QNH prediction, dewpoint forecasting, relative humidity, LightGBM, aviation meteorology

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TABLE OF CONTENTS

1 INTRODUCTION 1

1.1 Overall project context 1

1.2 Individual component context 1

1.3 Problem statement 3

1.4 Research gap 5

1.5 Aim and objectives 6

1.6 Research questions or hypotheses 7

1.7 Scope limitations and assumptions 8

1.8 Individual contribution 9

1.9 Thesis structure 10

2 LITERATURE REVIEW AND RELATED WORK 11

2.1 Review method and scope 11

2.2 Domain and theoretical foundations 12

2.3 Related research 14

2.4 Existing systems products and techniques 16

2.5 Gap synthesis and conceptual framework 18

3 RESEARCH METHODOLOGY 20

3.1 Research design 20

3.2 Data participants or study materials 21

3.3 Experimental or study procedure 23

3.4 Measures and evaluation metrics 26

3.5 Data analysis 27

3.6 Ethics privacy security and responsible AI 28

3.7 Reproducibility and research artefacts 29

4 PRODUCT AND SOLUTION DESIGN 30

4.1 Requirements 30

4.2 Component architecture within the complete solution 31

4.3 Design alternatives and decisions 33
4.4 Detailed design 34
4.5 Standards and professional practice 36
5 IMPLEMENTATION INTEGRATION AND TESTING 38
5.1 Development environment and technology stack 38
5.2 Component implementation 38
5.3 Integration with the other components 41
5.4 Verification and software testing 42
5.5 Deployment and operation 43
6 RESULTS AND VALIDATION 45
6.1 Research results 45
6.2 Product and technical results 46
6.3 User or stakeholder validation 47
6.4 Integration results 48
6.5 Summary of findings 49
7 DISCUSSION 50
7.1 Interpretation of findings 50
7.2 Comparison with literature baselines and existing solutions 51
7.3 Research contribution 52
7.4 Product contribution 52
7.5 Limitations and threats to validity 52
7.6 Implications 54
8 COMMERCIALISATION AND SUSTAINABILITY 55
8.1 Value proposition and target users 55
8.2 Market and competitive position 55
8.3 Business or sustainability model 56
8.4 Scalability, intellectual property and compliance 57
8.5 Contribution to the Sustainable Development Goals 57
9 CONCLUSION AND FUTURE DIRECTIONS 59

9.1 Conclusion 59
9.2 Achievement of objectives 60
9.3 Individual research and product contributions 60
9.4 Future directions 61
REFERENCES 62
GLOSSARY 64
APPENDICES 65
Appendix A Research instruments or extended results 65
Appendix B Academic integrity and AI use disclosure 65
Appendix C: Data dictionary and feature catalogue 67
Appendix D: End-to-end procedure and pseudocode 69
Appendix E: Evaluation protocol and metric definitions 72
Appendix F: Software and data-quality test matrix 75
Appendix G: Individual contribution and progress-log traceability 77
Appendix H: API and data-contract specification for review 79
Appendix I: Model card and intended-use statement 81
Appendix J: Reproducibility and final-submission checklist 83
Appendix K: Monsoon evaluation worksheet and error analysis 85
Appendix L: Operational risk register and runbook 86
Appendix M: RH equation notes and interpretation boundaries 88
Appendix N: Final result-reporting templates 89

LIST OF FIGURES

Figure 3.1. Feature engineering families described in the group research paper.....23

Figure 3.2. Research workflow from historical observations through training and verification.....24

Figure 4.1. Component architecture for QNH, dewpoint and RH prediction.....32

Figure 5.1. RH derivation pipeline using predicted temperature and dewpoint.....40

Figure 5.2. Forecast, verification and model-maintenance cycle.....44

Figure 6.1. Summary of reported QNH and dewpoint metrics.....46

LIST OF TABLES

Table 3.1. Dataset and preprocessing summary.....21

Table 4.1. Functional requirements of the individual component.....30

Table 6.1. Reported QNH and dewpoint evaluation results.....45

Table 9.1. Objective achievement and evidence summary.....60

Table A.1. Component verification checklist.....65

Table C.1. Core observation, forecast and verification fields.....67

Table C.2. Candidate engineered feature families.....67

Table E.1. Required evaluation artefacts.....72

Table E.2. Reported component metrics.....73

Table F.1. Software and data-quality test matrix.....75

Table G.1. Individual contribution and progress-log traceability.....77

Table H.1. Forecast response contract.....79

Table H.2. Endpoint inventory to reconcile with code.....79

Table I.1. Model card and intended-use summary.....81

Table J.1. Dataset record template.....83

Table J.2. Model record template.....83

Table J.3. RH reproducibility record.....83

Table K.1. Monsoon evaluation worksheet.....85

Table K.2. Error-case review form.....85

Table L.1. Operational risk register.....86

Table N.1. Overall result-reporting template.....89

Table N.2. Baseline comparison template.....89

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
DA	Density altitude
hPa	Hectopascal
ICAO	International Civil Aviation Organization
LightGBM	Light Gradient Boosting Machine
MAE	Mean absolute error
METAR	Meteorological aerodrome report
ML	Machine learning
MLOps	Machine learning operations
QNH	Altimeter setting based on atmospheric pressure reduced to mean sea level
R^2	Coefficient of determination
RH	Relative humidity
RMSE	Root mean squared error
TAF	Terminal aerodrome forecast
Td	Dewpoint temperature
T	Dry-bulb air temperature
UTC	Coordinated Universal Time
VCBI	Bandaranaike International Airport ICAO location indicator
FastAPI	Python web framework used for backend APIs
JSON	JavaScript Object Notation
REST	Representational State Transfer
UI	User Interface
CSV	Comma-Separated Values

1 INTRODUCTION

1.1 Overall project context

Weather is a critical operational input for civil aviation. Changes in atmospheric pressure, temperature, humidity, wind, cloud, visibility and precipitation can influence approach planning, runway use, flight dispatch and the assessment of operational risk. Airport-scale conditions may change more quickly than broad regional summaries convey, so short-term forecasts at a particular aerodrome can provide additional context for local decision-making. Aviation meteorology therefore benefits from forecasts that are both temporally current and tied to the location where operational decisions are made [4], [5], [14].

Sri Lanka experiences four commonly recognised seasonal regimes relevant to this study: the Southwest Monsoon, Northeast Monsoon, First Inter-Monsoon and Second Inter-Monsoon. The distribution and intensity of rainfall, humidity, wind and cloud can vary across these periods. A model trained on historical observations must therefore account for changes by hour and season and should be evaluated on later observations rather than randomly shuffled records. The supplied group research paper frames this problem around Bandaranaike International Airport (VCBI) and a hyper-local forecasting and verification system [16].

The wider project combines meteorological data acquisition, preprocessing, machine-learning forecasts, aviation-related calculations, validation against later observations and visual presentation of the results. It is modular: different components handle temperature and pressure, wind, cloud and visibility, and QNH, dewpoint and relative humidity. This thesis focuses only on the last component, while describing its interfaces with the complete system where necessary.

1.2 Individual component context

The individual component predicts QNH and dewpoint three hours into the future and derives relative humidity from the predicted dewpoint and a predicted dry-bulb temperature supplied by the wider forecasting system. QNH is a pressure value used in aviation altimetry. Dewpoint represents the temperature to which air must be

cooled, at approximately constant pressure and water-vapour content, for saturation to occur. Relative humidity expresses the degree of saturation as a percentage and depends on both air temperature and dewpoint.

The distinction between modelled and derived outputs is important. The supplied paper reports separate LightGBM models for QNH and dewpoint. RH is calculated with the August-Roche-Magnus relationship rather than learned as a third independent output. This design uses the predicted atmospheric variables to keep the humidity calculation physically interpretable. The RH estimate is only as reliable as the temperature and dewpoint values supplied to the equation, and its error should be evaluated separately before operational decisions are based on it [16].

The component operates within a system that stores historical observations and time-stamped forecasts, receives current METAR observations, and compares a forecast with an observation when the forecast target time is reached. It is consequently not just a model-training task: data preparation, forecast timestamps, consistent feature generation, persistence and post-event verification are part of the component lifecycle.

Operational meaning of the three component variables

The component treats QNH, dewpoint and relative humidity as related but not interchangeable quantities. QNH is a pressure setting expressed in hectopascals (hPa); dewpoint is a temperature expressed in degrees Celsius (°C); and relative humidity is a percentage describing how close the air is to saturation at the prevailing temperature. The modelling plan therefore has two learned forecast targets and one derived output. This separation avoids describing RH as an independently trained LightGBM output when the supplied paper instead calculates RH using predicted dewpoint and predicted dry-bulb temperature.

For aviation users, the value is not simply the production of three numbers. It is production of time-aligned estimates whose issue time, target time, source observations and model version can be identified. A pressure estimate with no target timestamp can be misread as a current measurement. A dewpoint estimate compared with a different hour can appear inaccurate or accurate for the wrong reason.

Calculated RH can also become physically inconsistent if the temperature and dewpoint forecasts refer to different target times. Timestamp consistency is therefore a design property, not an administrative detail.

Relationship to the wider group system

The group system is modular. The individual component consumes historical weather records and current meteorological observations, creates three-hour-ahead QNH and dewpoint forecasts, derives RH and makes the outputs available to the wider application and verification process. The group paper describes a common pipeline for acquisition, preprocessing, model inference, forecast storage, validation and visual presentation. This thesis documents only the QNH/dewpoint/RH component; it does not claim ownership of the other members' temperature, wind, cloud or visibility models.

The dependency on predicted temperature is especially important. The RH calculation uses the forecast temperature produced by the wider system together with the component's predicted dewpoint. When an aligned temperature forecast is not available, the correct behaviour is to mark RH unavailable or partial—not to substitute an unlabelled observation from another time. The final integration should record the source and target timestamp of the temperature value.

Component boundary and outputs

The boundary starts with weather observations and ends with timestamped forecast products and verification records. Internally, the component includes input checks, temporal ordering, missing-value treatment, feature calculation, model loading, inference, RH calculation, output validation and persistence. External interfaces supply or receive data but do not change the mathematical meaning of the outputs. A pressure or dewpoint model can produce a prediction successfully while the service still fails if the database write, response serialization or target timestamp is wrong.

1.3 Problem statement

General regional weather prediction does not necessarily provide the specific pressure, dewpoint and humidity estimates needed for a single airport over a short forecast horizon. At VCBI, forecast usefulness depends on a model learning local

temporal behaviour from historical data while accommodating monsoon-related changes. The task is challenging because consecutive observations are dependent, missing records can occur, and feature values used during inference must be constructed in the same way as values used during model training.

The individual research problem is to produce a reproducible three-hour-ahead QNH and dewpoint prediction workflow for VCBI, derive a corresponding RH estimate from physical relationships, and assess the predictions against subsequently observed meteorological values. An additional engineering challenge is to make the forecasts available to the broader system and retain their target timestamps so that performance can be checked over time. The thesis uses the results reported in the provided research paper; it does not introduce unreported experiments or raw observations.

Research problem decomposition

The problem can be separated into four connected questions. First, are historical observations ordered and cleaned in a way that preserves the sequence available at forecast issue time? Second, can learned relationships between weather variables and their earlier values provide useful three-hour-ahead pressure and dewpoint estimates? Third, can RH be derived using temperature and dewpoint values that refer to the same valid time? Fourth, can the system retain a forecast and later compare it with the observation that verifies that specific forecast?

These questions should not be collapsed into a single claim of accuracy. A model score reflects a defined target, data partition, metric and evaluation sample. Data quality affects whether the score is meaningful. Formula implementation affects RH consistency. Integration affects whether a user receives the correct result at the correct time. This thesis consequently considers scientific performance and software behaviour separately before discussing their combined value.

Why a three-hour horizon matters

The research paper identifies three hours as the main horizon for the QNH/dewpoint component. For an input record at time t, the target must correspond to the intended future time t+3 hours. The model should not accidentally train on the current value

while being described as a future predictor. The implementation and evaluation pipeline must make the shift between feature time and target time explicit.

The chosen horizon balances lead time and forecast difficulty. Short-horizon predictions can exploit temporal continuity, while abrupt changes and seasonal transitions can still reduce accuracy. The paper does not report a complete horizon-by-horizon comparison, so this thesis does not claim that three hours is mathematically optimal; it treats the horizon as the chosen scope of the system.

1.4 Research gap

Prior work cited by the group paper includes machine-learning approaches to monsoon rainfall and short-term weather prediction in Sri Lanka [1], [2], as well as broader research into neural-network weather forecasting, ensemble post-processing and sequence models [7]–[10]. Aviation-oriented studies in the same reference set discuss high-impact weather, airport-specific lightning climatology and automated terminal aerodrome forecasting [3]–[6], [13]. These works motivate applying prediction methods to weather variables that matter at an aerodrome.

The gap addressed by the project is the integration of local pressure and dewpoint forecasting with a physically derived RH output and a future-observation verification loop for VCBI under Sri Lankan monsoon conditions. The provided literature review identifies that existing Sri Lankan studies largely focus on general weather or monsoon rainfall, rather than the joint airport-specific treatment of these variables. The thesis component operationalises this gap through time-aware preprocessing, separate forecasts and a traceable forecast-to-observation comparison [16].

Specific gap addressed by the individual study

The project addresses a gap at the intersection of local pressure forecasting, local dewpoint forecasting and a physically based RH estimate embedded in an airport-specific verification workflow. Prior Sri Lankan work cited in the group paper addresses monsoon rainfall or general short-term weather prediction [1], [2]. Other sources discuss aviation weather hazards, automated terminal forecasts, deep-learning approaches and the role of data assimilation [3]–[13]. These sources provide relevant foundations, but the paper identifies no directly matching study that jointly

implements the described QNH, dewpoint and derived-RH workflow for VCBI under Sri Lankan monsoon conditions.

The gap is not that no weather forecasting systems exist. The narrower contribution is the integration and evaluation of a specific component in a local application. LightGBM is not presented as a novel algorithm. The research contribution lies in adapting a model to a defined airport/time-series task, documenting how dependent RH is derived, and making forecasts available for time-aligned comparison with later METAR observations.

The novelty claim must remain proportional to the evidence. The research paper contains summary metrics but not a full experiment register, all trained model files, season-wise evaluation tables or a separate RH accuracy result. The thesis therefore frames the contribution as an implemented and evaluated component with reproducibility gaps to be closed, rather than a fully demonstrated all-season forecasting breakthrough.

1.5 Aim and objectives

The aim of the individual component is to develop and evaluate a data-driven QNH and dewpoint forecasting module and a physically based RH calculation for VCBI, for integration into the wider hyper-local aviation weather forecasting and verification system.

1. Develop a chronological preprocessing workflow for historical VCBI weather records, including missing-value handling and time-dependent feature creation.
2. Train separate LightGBM models to predict QNH and dewpoint three hours ahead using the available historical observations.
3. Calculate forecast RH from predicted dry-bulb temperature and dewpoint using the August-Roche-Magnus relationship.
4. Evaluate QNH and dewpoint predictions using the reported error and fit metrics, and document which RH validation evidence is still required.
5. Integrate the component’s outputs into the wider persistence, API and forecast-verification workflow.

6. Document operational limitations and identify the additional validation needed before the outputs can be treated as aviation decision-support evidence.

Objective-to-evidence mapping

Every objective should connect to evidence that demonstrates completion. The paper supports the model family, forecast horizon, RH equation and headline dewpoint/QNH metrics. The progress log supports a dated record of implementation and integration work. It does not contain the full prediction arrays or test-run logs needed to independently recompute the reported metrics.

Preprocessing is supported by the paper's chronological-ordering and feature-engineering description plus the progress entries concerning cleaning. The QNH/dewpoint model objective is supported by the separate LightGBM three-hour models and summary scores. The RH objective is supported by the August 21 log entry recording implementation of the August-Roche-Magnus formula using predicted dewpoint and temperature. Evaluation is only partially evidenced: exact sample counts, test dates and a separate RH evaluation are still required. MLOps, manual inference and live-verification work are supported by the progress log, but a formal deployment report remains necessary.

1.6 Research questions or hypotheses

1. How accurately can a LightGBM model forecast QNH at VCBI three hours ahead using historical meteorological observations and time-derived features?
2. How accurately can the same general modelling approach forecast dewpoint three hours ahead, as measured by MAE and R^2?
3. How can predicted dewpoint and temperature be combined to derive an interpretable RH estimate?
4. How can time-stamped forecast storage and comparison with later METAR observations support ongoing model verification?
5. What limitations remain when the reported results are used to infer performance across different monsoon periods?

The supplied paper is an applied system study rather than a pre-registered hypothesis test. Accordingly, this thesis states research questions and evaluates them against the numerical evidence available in the paper; it does not claim statistical significance or seasonal generalisation that has not been reported.

1.7 Scope limitations and assumptions

The geographical scope is VCBI. The reported historical dataset spans 2019–2026 and contains more than 120,000 samples for the overall forecasting system. The individual component focuses on QNH and dewpoint prediction with a three-hour horizon and on calculating RH from predicted temperature and dewpoint. It also covers the described data pipeline, model evaluation and integration to the extent documented by the group paper [16].

The thesis does not claim to certify an aviation weather product or replace professional meteorological assessment. It assumes that historical observations and METAR records are sufficiently representative for model development, that timestamp alignment is correct, and that training and inference features are consistent. The provided paper does not specify the exact missing-value imputation rule, per-season sample counts, final holdout duration, code version, model hyperparameters for LightGBM, or a separate RH error score. Those omissions limit reproducibility and are treated as limitations rather than filled with invented values.

The group paper includes results from several other modules. Those results are discussed only when needed to describe system integration; they are not attributed to the individual QNH/RH/dewpoint component.

Assumption register

The study assumes that historical and METAR timestamps are normalised to a consistent time basis before feature generation and verification. The paper states that data are ordered using UTC times; the actual database schema should confirm this for all component paths. Inputs used at inference must be available at forecast issue time. Missing-value treatment must not use future observations to fill earlier features. The RH equation must receive temperature and dewpoint in degrees Celsius, aligned

to the same valid time. Finally, test metrics must be calculated against observations not used to fit the model.

These assumptions are not all independently demonstrated in the supplied paper. They are recorded as verification obligations. During finalisation, each should be confirmed from source code, configuration, the database schema or run outputs. If a check fails, the corresponding claim must be revised rather than hidden in a limitations paragraph.

Explicit out-of-scope items

The thesis does not establish a certified aviation meteorological service, replace the judgement of trained meteorological or flight-operations personnel, or independently verify every variable in the group project. It does not claim skill at airports other than VCBI, equal performance in every monsoon season, or RH accuracy based only on the equation's implementation. Calculated RH requires its own target-time comparison.

1.8 Individual contribution

The individual contribution reported by the student covers the component from data preparation through integration. This includes preprocessing historical meteorological records, creating time-based and rolling features, preparing a chronological training and test split, training the LightGBM QNH and dewpoint models, implementing the RH calculation, evaluating the reported model outputs, and connecting the component to the backend workflow. The student's component contribution is distinguished from other group members' work on wind, cloud and visibility, and temperature/pressure subsystems.

The supplied paper reports the headline metrics used in this draft, but it does not include the component's full code, a complete feature schema, an exact run log, a holdout size, or all model configuration details. These artefacts should be attached or referenced in the final thesis if they are available. The contribution statement should also be checked against the project repository and supervisor feedback before final submission.

Evidence basis for individual contribution

The progress log makes the work traceable. Early entries document a Random Forest regressor for QNH, dashboard and pipeline work, multi-output backend support, API/frontend structure and training/verification scripts. Later entries record reviewing additional historical data, monsoon-pattern analysis, data-cleaning strategies, model-family comparison, feature selection, API/database design, MLOps planning and final LightGBM three-hour models. August entries specifically record an auto-retraining utility, manual inference and live verification scripts, feature-selection logic inside preprocessing and RH formula implementation.

The log documents iteration rather than one uninterrupted implementation. Early Random Forest and gradient-boosting scripts were later cleaned up or removed, the focus was streamlined to QNH/dewpoint, and the project structure was aligned with the group branch. The final thesis should describe the latest supported configuration as the final design, while identifying superseded models as development experiments.

The student reports responsibility for preprocessing, LightGBM training, RH formula implementation, model evaluation and backend development. This contribution statement should be reviewed against the code repository and commit history so that implementation ownership is accurate and group-developed interfaces are not attributed exclusively to one member.

1.9 Thesis structure

Chapter 1 introduces the project, individual problem, objectives, research questions, scope and contribution. Chapter 2 reviews relevant literature and technical foundations; Chapter 3 presents the methodology and evaluation. Chapters 4 and 5 describe the component design, implementation and integration, while Chapter 6 reports the supplied results. Chapter 7 discusses their meaning and limitations, Chapter 8 considers potential sustainability, and Chapter 9 concludes and identifies future work. References and appendices provide source details and supporting evidence.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This review is based on the literature cited in the supplied group research paper. The paper covers machine learning for general weather forecasting, aviation weather forecasting, machine learning in aviation prediction, aviation hazards and the identified airport-level gap. It is therefore treated as a focused narrative review, not as a systematic literature review. No separate database search log, inclusion/exclusion procedure or formal quality-appraisal matrix was provided in the source material.

The scope relevant to this individual thesis is intentionally narrower than the full project: airport-scale time-series prediction, pressure and dewpoint as meteorological variables, computation of relative humidity from temperature and dewpoint, temporal feature engineering, future-observation verification, and the implications for aviation decision support. Research about flight-delay prediction or other aviation uses of machine learning is included only as contextual evidence that data-driven approaches have been applied in aviation; it is not treated as direct evidence of meteorological forecasting accuracy [11].

For the final submission, the reference metadata and in-text numbering should be cross-checked against the original publications. The supplied group paper contains several in-text citation-number inconsistencies relative to its reference list. This draft cites the entries by their order in the supplied reference list where author and title information are available, but the bibliography should be verified against the source publications before submission.

Screening and synthesis approach

The review is organised around the component's engineering decisions rather than as a general list of AI applications. Sources are relevant when they explain time-aware weather prediction, use of ML in Sri Lankan or aviation weather, airport-local verification or limitations of models in changing atmospheric regimes. The group paper supplies the source set; this is a focused narrative review, not a systematic

review with database searches, duplicate-screening counts or a formal PRISMA protocol.

The synthesis distinguishes direct evidence from transferable context. Sri Lankan rainfall and short-term weather studies provide geographical and methodological context, but their targets differ from airport-level QNH or dewpoint. Aviation-hazard studies motivate airport-specific meteorology, but do not directly validate this component's forecast error. Other ML studies show candidate techniques, but their performance cannot be transferred numerically to VCBI without common datasets, horizons and metrics.

For final submission, each reference should be checked against the original publication, including author spelling, title, venue, year, pages and identifier where available. The supplied group paper has some citation-number inconsistencies, so the final bibliography should be normalised consistently before final renumbering. This draft does not invent missing bibliographic details.

2.2 Domain and theoretical foundations

QNH is an aviation pressure setting derived by reducing observed atmospheric pressure to mean sea level under standard assumptions. It is commonly expressed in hectopascals and is important in altimeter setting. A pressure forecast can therefore support contextual interpretation of atmospheric conditions, although a forecast value on its own is not an operational clearance or safety determination. Its usefulness depends on the accuracy of the underlying observation, the forecast horizon and the procedures of the aviation organisation using it [14], [15].

Dewpoint is a measure of atmospheric moisture. When air temperature approaches dewpoint, relative humidity approaches saturation, and the chance of fog, low cloud or condensation can become operationally relevant depending on local conditions. Dewpoint is not interchangeable with RH: RH changes with temperature even when the amount of water vapour is unchanged. For this reason, separately predicting dewpoint and combining it with a temperature prediction can provide a coherent physical route to RH.

The component uses the August-Roche-Magnus approximation. Let T be predicted dry-bulb temperature in degrees Celsius and Td be predicted dewpoint in degrees Celsius. RH is then estimated as: RH = 100 × exp[(17.67 Td / (Td + 243.5)) − (17.67 T / (T + 243.5))]. The expression returns a percentage estimate. Its inputs must use the same temperature units and appropriate timestamps. A small error in either input can change the RH estimate; therefore, the model's temperature and dewpoint errors jointly influence the derived result.

LightGBM is a gradient-boosting framework designed for tree-based learning on tabular data. In this component, it is applied to lagged observations and engineered variables rather than to raw image or radar sequences. Time-dependent features can represent daily and annual periodicity, recent changes and short rolling trends. The chronological split protects the temporal ordering of the evaluation and is more appropriate than randomly shuffling serial observations for a short-horizon forecasting task [9], [16].

Time-series learning and data leakage

A time-series forecasting task differs from random independent-row prediction because observation order carries information. The model should learn from records available at a given issue time and predict a later target. If rows are randomly shuffled before splitting, later records can enter training while earlier or nearby records appear in testing. That split no longer represents training on the past and predicting the future. The group paper specifies chronological training and testing without shuffling, which should be preserved for these models.

Leakage can also occur inside preprocessing. Examples include rolling averages that include future records, fitting a scaler on the full dataset before splitting, filling a past missing value using a later observation, or joining an input row to the wrong target timestamp. A safe pattern defines issue-time features and the shifted target, creates chronological partitions, fits learned transformations on training data, and applies them to later partitions. The actual implementation should be inspected to confirm these controls.

Gradient-boosted decision trees as a modelling choice

LightGBM is a gradient-boosted decision-tree method. In broad terms, boosting builds an ensemble in stages, with later trees reducing residual error under a chosen objective. Tree-based methods can represent nonlinear interactions between lagged weather variables, time indicators and engineered differences without requiring every relationship to be specified as a linear equation. This makes them a reasonable candidate for structured tabular meteorological records.

A model family alone does not guarantee accurate forecasts. Performance depends on the target definition, feature quality, training-data representativeness, hyperparameters, evaluation period and baseline strength. The group paper reports LightGBM for QNH and dewpoint but does not provide the full model configuration. The thesis can explain the method choice but should not claim that its hyperparameters are optimal without tuning logs.

QNH, dewpoint and RH as a linked physical system

QNH and dewpoint are distinct targets with different physical meanings and units. Pressure error should be expressed in hPa, dewpoint error in degrees Celsius, and RH error in percentage points. These errors should not be combined or directly compared without a clearly defined normalisation scheme. RH depends on both temperature and dewpoint; input errors propagate into the derived value. The equation is interpretable but does not make the output error-free.

Why verification is part of the theoretical design

Forecast verification links a prediction to an observation available at the corresponding valid time. This closes the loop between development and actual performance. The project paper describes saving forecasts with target timestamps and later matching them with METAR reports. This allows the system to distinguish what was predicted from what was observed and to identify when the comparison was made.

2.3 Related research

For the Sri Lankan context, Nagahamulla, Ratnayake and Ratnaweera studied artificial-neural-network methods for monsoon rainfall forecasting, while Hennayake

and colleagues reported a machine-learning model for short-term weather prediction in Sri Lanka [1], [2]. These works demonstrate local interest in data-driven meteorological prediction. Their focus, as described in the supplied paper, is not the same as the airport-level QNH, dewpoint and RH workflow addressed here.

Research on sequence models also motivates the use of temporally structured information. Shi and colleagues introduced ConvLSTM for precipitation nowcasting, and Rasp and Lerch studied neural-network post-processing of ensemble forecasts [7], [10]. Dueben and Bauer discuss design challenges for machine-learning weather and climate models [9]. These studies provide broader methodological context but do not establish the performance of the specific VCBI LightGBM models used in this component.

Aviation-specific literature in the supplied list covers the climatology of lightning activity over VCBI, high-impact weather for aviation meteorology, automated terminal aerodrome forecasting and lightning-risk prediction [3], [4], [6], [13]. The reference set also includes a discussion of common aviation weather hazards [5], and the role of suitable observations in numerical weather prediction is represented by data-assimilation research [12]. These sources reinforce the value of airport-specific observations and the need to interpret weather forecasts in an aviation context.

The cited research supports the rationale for the component but should not be interpreted as a direct performance comparison. The current thesis has only the reported QNH and dewpoint metrics in the group paper, not a matched benchmark dataset evaluated with identical features, target times and chronological test windows. A controlled comparison with persistence, a simple climatological baseline and alternative regression algorithms would strengthen the evidence.

Sri Lankan weather prediction studies

Nagamulla, Rathnayake and Ratnaweera investigate monsoon-rainfall forecasting in Sri Lanka using artificial neural networks [1]. Hennayake and colleagues address short-term weather prediction using machine-learning methods and meteorological variables [2]. These are relevant because they show local interest in data-driven weather forecasting, but rainfall and general weather are not identical targets to QNH

or dewpoint. Comparisons should remain conceptual unless datasets, horizons and metrics align.

The transfer to the current component is methodological: local historical observations can contain patterns useful to a predictive model, but local preprocessing and evaluation are essential. The monsoon setting motivates season-aware evaluation. An aggregate test score may hide weaknesses in a specific seasonal regime or during abrupt changes. The source paper provides the motivation but not a complete seasonal performance table.

Aviation meteorology and airport-specific information

The aviation meteorology review and other aviation-hazard studies cited by the group paper support the need for airport-specific weather information [3]–[6], [13]. They do not establish that this project's QNH/dewpoint forecasts meet a formal aviation threshold. An airport-level prototype can offer contextual information and support review, but operational adoption requires formal performance monitoring, stakeholder review and compliance checks.

Machine-learning techniques and feature design

The cited literature includes neural networks, ConvLSTM, AI weather prediction and ensemble post-processing [7]–[10]. Sequence-based neural models can represent temporal dependencies explicitly, whereas tree ensembles often benefit from engineered lags, rolling statistics and cyclic time variables. The individual component uses a tree-boosting design combined with temporal features and a physical RH equation. This is a pragmatic integration, not a claim that LightGBM dominates other methods.

2.4 Existing systems products and techniques

Traditional aviation weather products include METAR observations and TAF forecasts. METAR provides a coded report of observed aerodrome weather; TAF provides a forecast over a defined period. In the group system, METAR observations from the Aviation Weather Center are also used as machine-readable inputs and as later observations for checking predictions. This gives the component a practical

verification reference, while the data still require parsing, quality control and accurate time alignment [15], [16].

Statistical and machine-learning techniques have different trade-offs. A persistence forecast is simple and useful as a baseline but may miss rapid changes. Linear regression offers interpretability but may be too restrictive for nonlinear interactions. Tree ensembles, including LightGBM, can represent nonlinear relationships among lagged weather variables and engineered features. Sequence models may learn longer temporal dependencies but require additional training choices and careful validation. The supplied research paper selected LightGBM for QNH and dewpoint; it does not provide a comparative experiment of alternative algorithms for those two targets, so the selection should be described as the implemented choice rather than a proven universal optimum.

Commercial weather platforms and national meteorological services may provide general forecasts and aviation weather information, but the supplied paper does not present a product-by-product market evaluation. This thesis therefore compares techniques conceptually and avoids unsupported claims about competitor accuracy, market share or operational certification.

Comparison dimensions for existing systems

A fair comparison with an existing product needs the same valid time, forecast horizon, station, variable, unit, data availability and evaluation dates. Comparing a three-hour model with a six-hour forecast mixes horizon effects with model effects. A pressure convention mismatch can also create a false error. Therefore this thesis discusses existing techniques in terms of capability and limitations instead of unsupported numerical superiority.

A useful benchmark set would include a persistence baseline and alternative ML models assessed on the same chronological test split. The progress log records that Random Forest and XGBoost were examined, but the final paper does not present a complete QNH/dewpoint comparison for these models on identical cases. Historical exploration must not be converted into a claim that LightGBM outperformed all alternatives unless those results are supplied.

Operational product comparison

General weather applications, aviation forecast products and this hyper-local research component may have different authority, coverage, refresh frequency and validation standards. A future comparison should document each product's variables, spatial scale, forecast horizon, provenance, update policy and whether the score uses the same observations. The current source set does not contain a complete market or product benchmark.

2.5 Gap synthesis and conceptual framework

The reviewed work supports four requirements: local observations are necessary for a local forecast; temporal order must be respected; predicted meteorological variables should be evaluated against later observations; and aviation outputs must be presented with their uncertainty and limitations. The remaining gap is addressed by joining three functions in one individual component: three-hour QNH prediction, three-hour dewpoint prediction, and formula-based RH derivation from predicted dewpoint and temperature.

The conceptual workflow is: collect and persist historical observations; order records by UTC; handle missing values; derive lagged, cyclic-time and rolling features; fit separate LightGBM regressors; generate forecasts for the target timestamp; calculate RH from predicted temperature and dewpoint; store outputs; then match forecasts to future METAR observations when available. This structure makes it possible to trace each forecast to its input window and its eventual verification result. It also exposes where audit artefacts are needed, including an explicit imputation strategy, model configuration, data split boundaries and separate RH evaluation.

Conceptual model of the individual component

The framework consists of evidence-preserving transformations. Historical observations are converted into cleaned time-series records. Issue-time features are calculated using information intended to be available at that time. Two regression models estimate future QNH and dewpoint. Predicted dewpoint is combined with a temperature forecast through the August-Roche-Magnus relationship to derive RH. Outputs are stored with issue time, target time and model/component identifiers.

When the target-time observation becomes available, the forecast is matched with it and verification metrics can be calculated.

Each stage has a distinct failure mode: timestamp errors can shift targets; unsuitable imputation can distort patterns; leakage can inflate test scores; model drift can reduce future performance; inconsistent inputs can distort RH; and incorrect matching can corrupt verification. Robustness therefore requires controls across the pipeline, not only a high model score.

Claims supported and not supported by the source

The source supports the use of historical VCBI meteorological data, a 2019–2026 period and more than 120,000 samples for the wider system, three-hour QNH/dewpoint forecasting using LightGBM, and RH calculation from predicted temperature and dewpoint. Reported component metrics are dewpoint $R^2 \approx 0.66$ and MAE ≈ 0.58 °C, and QNH MAE ≈ 1.8 hPa. A separate RH score, exact individual sample count, full model configuration and seasonal results are not supplied.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows an applied, quantitative model-development and evaluation design. Historical observations are used to estimate the relationship between recent meteorological conditions and future QNH/dewpoint values. The trained models produce three-hour-ahead predictions, and those predictions are compared with observed values at the target time. The group paper describes a modular system and uses time-stamped forecasts for subsequent verification, which is consistent with an operational forecasting workflow rather than a cross-sectional analysis [16].

The approach is retrospective for model development and operationally oriented for inference. The reported 80/20 chronological train-test division avoids mixing future samples into the training data. However, the source paper does not report the precise start/end timestamps of the holdout, the number of records in each split, or separate results by monsoon. Those details are important to make the evaluation independently reproducible and should be added from the actual training logs if available.

Research design rationale

The work follows an applied, artefact-oriented research design. Its output is both a forecasting method and an implemented software component. The research question is addressed through component development, comparison with later observations where evidence is available, and analysis of engineering constraints. The sources do not describe a controlled human-participant experiment. Primary evidence includes meteorological time-series data, software artefacts, model outputs and the progress record.

The design separates development data from evaluation data and model quality from integration quality. Development includes feature choices and model fitting. Evaluation measures forecasts on a chronological period not used to fit the model. Integration tests check APIs, storage, timestamps and formula handling. A successful API response does not prove predictive accuracy; a good retrospective metric does not prove reliable live delivery.

Unit of analysis and forecast definition

The unit of analysis is a forecast case for VCBI and a specified valid time. Each case should identify issue time, target time, input features, predicted QNH, predicted dewpoint, temperature input used for RH, derived RH and later observations used for verification. If multiple forecasts target the same valid time, they should remain distinguishable by issue time and model version.

The target shift must be consistent. For an input observation at time t, the target is the value at t+3 hours. The sampling interval and missing-hour handling determine whether a fixed row offset equals three elapsed hours. Timestamp-based alignment or confirmation of regular sampling is therefore important.

3.2 Data participants or study materials

The primary study material is historical VCBI meteorological data reported to cover 2019–2026 and more than 120,000 samples for the overall system. The QNH/dewpoint subsystem uses local weather data stored in SQLite. The wider system also receives current METAR observations from the Aviation Weather Center and stores observations, predictions and verification logs using Supabase and SQLite. No human participants are described in the supplied research paper; the work is based on meteorological records [16].

Table 3.1. Dataset and preprocessing summary

Aspect	Information described in the research paper	Purpose in the component
Historical period	2019–2026; more than 120,000 samples for the overall system.	Training and chronological evaluation.
Location / source	VCBI historical meteorological observations.	Airport-specific modelling.
Live observations	METAR obtained from the Aviation Weather Center.	Current inference and future-observation verification.
Time handling	Ordered by UTC; forecast target shifted by three hours.	Temporal consistency and T+3 prediction.
Feature families	Cyclic hour/year; temperature-dewpoint difference; QNH movement; 3 h and 6 h rolling averages.	Represent periodicity and recent trends.
Split / storage	80% train and 20% test chronologically without shuffling; SQLite and wider Supabase integration.	Reduce temporal leakage and support persistence.

The source indicates that the data are ordered by UTC timestamps. It does not provide a complete data dictionary, source-by-source record counts, sensor provenance for every field, or the exact treatment of duplicate METAR messages.

The final thesis should attach the column schema and data-source description if those artefacts are available. At a minimum, each record used for training must identify observation time, target variables, predictor variables and the future target timestamp produced by the three-hour shift.

Data provenance and record inventory

The research paper describes historical VCBI meteorological data covering 2019–2026 and more than 120,000 samples for the wider system. It also describes current METAR observations from the Aviation Weather Center, with some records stored in SQLite and parts of the wider system using a cloud database service. The final thesis should name the actual file/table for each component and list source, timestamp range, sampling frequency, variables, units, missingness and access method.

The progress log says an additional 2024–2026 dataset was identified and reviewed in June 2026, and that missing-value strategies were considered. This documents data exploration but does not prove every added record entered the final training run. Inclusion should be confirmed from the extraction query or training manifest.

Variables and units

QNH is described in hPa and temperature/dewpoint in degrees Celsius; RH is a percentage. The dataset inventory should declare those units consistently. If raw feeds use another encoding, conversion should be documented. UTC timestamps should remain canonical for training and verification, while local-time conversion should occur only for display.

A numeric data type does not guarantee a valid weather value. Plausible-range checks and schema validation can catch unit mismatches, strings where numbers are expected, and humidity values outside the expected percentage range. The exact limits should be selected from the source convention and project requirements, not guessed. The feature families described for the broader pipeline are summarised in Figure 3.1; confirm the final subsystem-specific feature list against the saved code.

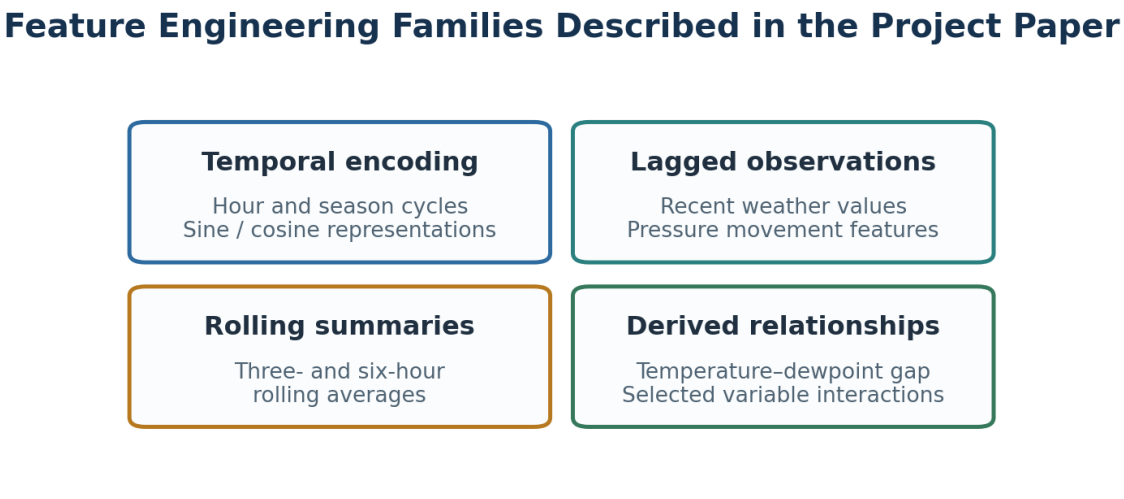


Figure 3.1. Feature engineering families described in the group research paper. The exact QNH/dewpoint feature subset should be confirmed against code.

3.3 Experimental or study procedure

1. Load the historical VCBI weather observations and preserve the original timestamps and target values.
2. Order observations chronologically using UTC and apply the recorded quality-control steps, including duplicate handling where applicable and missing-value processing.
3. Construct predictors from recent weather values and time metadata. The paper describes hour-of-day and time-of-year sine/cosine features, temperature-dewpoint difference, QNH movement and three-hour and six-hour rolling averages.
4. Create the forecast targets by shifting QNH and dewpoint three hours into the future. Keep the temporal sequence intact and split the records chronologically into 80% training data and 20% test data.
5. Train separate LightGBM regression models for QNH and dewpoint on the training partition, then generate predictions for the chronological test period.
6. Calculate RH using the predicted dewpoint and the predicted dry-bulb temperature from the wider system, ensuring that the input predictions refer to the same target timestamp.

7. Save forecasts with target timestamps and compare them with subsequent METAR observations once those observations become available.
8. Summarise the reported metrics and document any missing evidence rather than replacing it with estimated or simulated results. The full sequence is summarised in Figure 3.2.

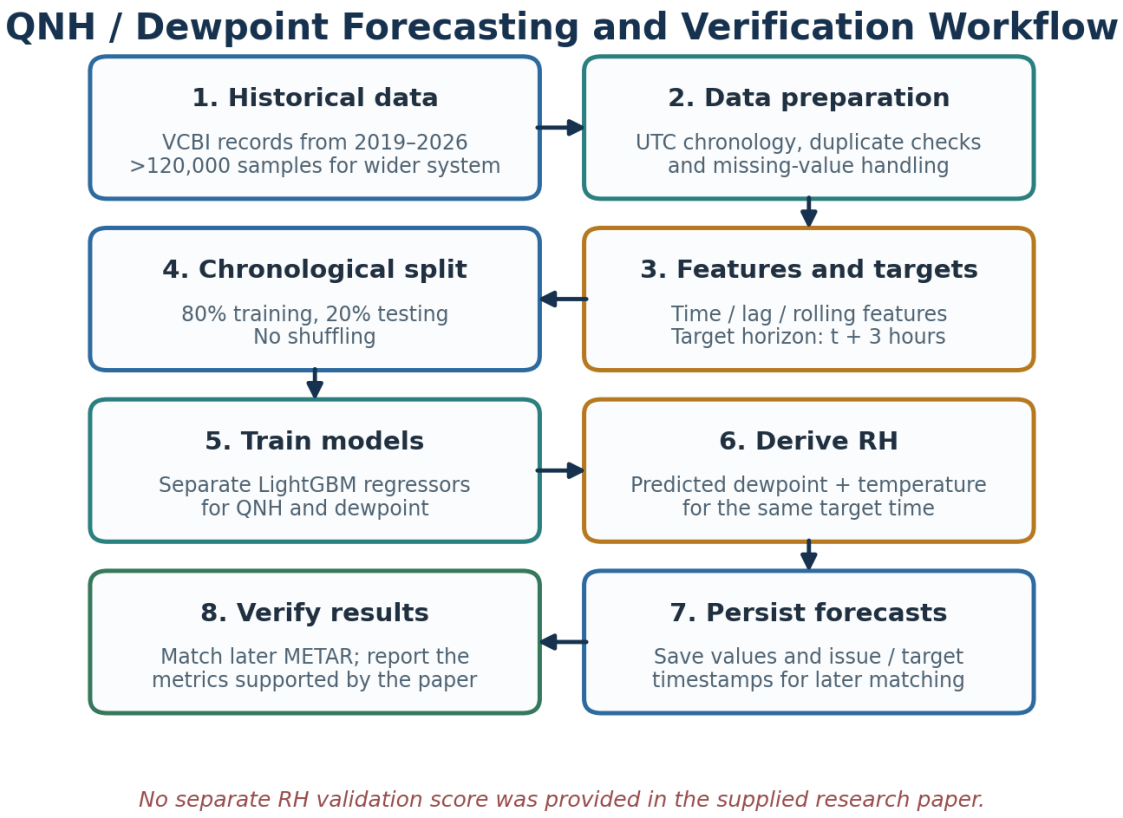


Figure 3.2. Research workflow from historical observations through LightGBM training, RH derivation and later METAR verification.

The source paper does not give LightGBM hyperparameters, random seed, software versions or an explicit missing-value imputation formula. Those omissions are not inferred here. For reproducibility, the implementation should store these items in configuration files or experiment logs and include them in the submitted supplementary material.

Detailed preprocessing sequence

First, parse and validate timestamps, convert records to the canonical time basis, sort them and inspect duplicates. Duplicates require a documented policy such as retaining the canonical observation or quarantining conflicts. Second, inspect missingness by variable and period. Missing-value treatment should not leak future information into past features. Third, construct the features actually used by the individual component. The group paper lists several feature families for the full system, but only features confirmed in the QNH/dewpoint pipeline should be claimed as active inputs.

Training and test separation

The paper reports an 80/20 chronological split for the subsystem without shuffling. A reproducibility record should additionally give the date boundary and number of valid cases for each target. Hyperparameter selection, if used, should occur inside the training period rather than repeatedly tuning against the final holdout. Learned transformations should be fitted on the training partition and reused for the test data.

Tree-based models may not require scaling in every configuration, but the paper says Min-Max normalisation was applied where required in the wider system. This thesis should describe what the component actually did instead of assuming that scaling is always necessary or never used.

Inference and verification lifecycle

At runtime, a valid input record is fetched, checked and transformed by the shared feature pipeline. The model produces the forecast and the system attaches a target time. RH is calculated only after a temperature forecast for the same valid time is available. Forecasts should be saved before verification so that the future observation does not overwrite the original prediction.

When a corresponding METAR observation becomes available, the verification routine matches by target time using a documented rule. The matched observation, forecast, time difference and metric inputs should be retained. If no valid observation exists, the forecast should remain unverified and be counted as such rather than silently omitted.

3.4 Measures and evaluation metrics

Mean absolute error (MAE) is the mean absolute difference between prediction and observation. It is useful because it remains in the unit of the predicted variable: hPa for QNH and degrees Celsius for dewpoint. The coefficient of determination (R^2) indicates how well the predictions account for variation in the target relative to a baseline based on the test-set mean, under the standard definition. Root mean squared error (RMSE) is discussed as a general metric in the group framework, but the supplied component results do not include an RMSE value for QNH or dewpoint.

$$\mathbf{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|$$

$$R^2 = 1 - \left[\sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2\right]$$

The paper reports dewpoint MAE and R^2 and QNH MAE. It does not report a separate RH MAE, RMSE or other validation score. Because RH is derived from two predicted inputs, it should be evaluated by computing RH from the predicted temperature and dewpoint and comparing that value with RH derived from, or directly observed alongside, the target-time meteorological record. Any such evaluation must use a consistent formula and timestamp alignment.

Interpretation of error metrics

Mean Absolute Error is the mean absolute difference between forecasts and observations, expressed in the target unit. The reported dewpoint MAE of about 0.58 °C is an average absolute discrepancy for the paper's evaluation sample; it does not mean every forecast is within that distance or that no large errors occurred.

Root Mean Squared Error takes the square root of the mean squared error, giving relatively greater influence to large misses. The coefficient of determination R^2 compares residual variation with variation around the observed mean. A value near 0.66 is not a 66% accuracy score and does not mean 66% of predictions are correct. The paper mentions MAE, RMSE and R^2 generally, but the individual QNH/dewpoint summary provides only dewpoint MAE/R^2 and QNH MAE.

Separate metrics for separate targets

QNH, dewpoint and RH must be reported separately because their units differ. Pressure error belongs in hPa, dewpoint error in °C and RH error in percentage

points. For RH, the model-derived value must be compared with a target-time reference using a consistent definition. Until this is performed, the thesis can document the formula implementation but cannot claim a measured RH accuracy.

3.5 Data analysis

The analysis compares predicted targets with their corresponding future observations and summarises model error. For the individual component, this means interpreting QNH error in hPa and dewpoint error in degrees Celsius, alongside the fit statistic available for dewpoint. Forecast timestamp and observation timestamp are key join fields; if those are misaligned, the computed error will not measure the intended forecast horizon.

The reported paper values are treated as summary outcomes, not as a new independent re-analysis of the source dataset, because the raw samples and prediction files were not supplied with the paper. Consequently, confidence intervals, per-month metrics, residual plots and formal significance tests cannot be reproduced from the current materials. They are appropriate follow-up analyses once the underlying forecast/observation table is available.

A stronger analysis would report results for each monsoon regime, dry versus wet periods, large pressure changes and humidity conditions near saturation. Such stratification could reveal error patterns masked by an overall mean. It should be designed before inspecting the final holdout results where possible to reduce selective reporting.

Descriptive and diagnostic analysis

Before fitting, data analysis should summarise each target’s valid record count, missingness, minimum and maximum, central tendency, spread and time coverage. Visual inspection should look for discontinuities, repeated values, unit problems and longer gaps. Unusual values should be checked against source records rather than automatically removed.

After forecasts are joined to observations, residuals should be calculated with a clear sign convention. Residual plots over time can reveal bias or changes in behaviour. Error distributions can show whether low MAE coexists with occasional large errors.

Seasonal and monthly analysis is a future validation need because aggregate scores can obscure regime-specific weaknesses.

Avoiding unsupported significance claims

The supplied report does not provide confidence intervals, formal significance tests or prediction-by-prediction data. This thesis therefore reports summary metrics without statistical-significance claims. A later comparison should use identical test cases and report sample counts and uncertainty where feasible.

3.6 Ethics privacy security and responsible AI

The described dataset consists of weather observations rather than personal data, and no human-subject study is described in the supplied paper. Even so, data-use permissions, attribution to the provider, and compliance with the terms of the Aviation Weather Center source should be retained with the project records. Any institutional ethics determination or data-access approval should be stated only if it was actually obtained; none is documented in the materials provided for this draft.

Weather predictions used in aviation are safety-relevant decision-support information. The component should therefore be presented as supplementary analytical output, not as a replacement for official METAR/TAF products, qualified meteorological judgement or approved operational procedures. Warning labels should identify prediction time, target time, units and data freshness. When data are missing or stale, the interface should avoid displaying an apparently current prediction without an explicit status.

The final implementation should protect API credentials, restrict write access to persisted observations and forecasts, preserve logs needed to trace a result, and avoid exposing internal service secrets to the frontend. These are recommended responsible-engineering controls; the supplied paper does not describe a security audit or provide evidence of specific access-control tests. Their implementation status should be verified against the actual code before claiming compliance.

Responsible presentation of forecasts

Users can mistake a prediction for an observation if the interface does not distinguish them. The application should show issue time, target time, source, freshness and

verification status. RH should be labelled as calculated from predicted temperature and dewpoint, not as a separately observed variable or an independently trained model.

The system should avoid false precision. Values should be displayed to a justified precision and should not imply certification. When the source is stale or a model is unavailable, the application should display an explicit unavailable state rather than reusing a previous forecast as if it were current.

3.7 Reproducibility and research artefacts

Reproducibility requires more than a description of the algorithm. The submitted artefacts should ideally include a data dictionary; the timestamp and missing-value rules; feature-generation code; exact training/test boundaries; LightGBM parameters; dependency versions; saved model identifiers; target horizon; evaluation script; and the output table used to compute metrics. If any source records cannot be redistributed, a schema and reproducible preprocessing script can still clarify the experiment.

The group paper indicates an SQLite data store and a wider Supabase-backed live system, plus a FastAPI backend. Model retraining and validation are described as recurring operational tasks. The final thesis should identify which scripts implement these steps and whether the results in Chapter 6 came from a specific saved model version. Since those repository artefacts were not included with the paper, this draft does not invent script names, versions or exact configuration values.

Minimum reproducibility bundle

A reproducible release should include a source description, feature definitions, target-shift rule, train/test boundaries, preprocessing configuration, model parameters, dependency versions, evaluation outputs, model version/checksum, code revision and training timestamp. Missing artefacts should be listed explicitly. The progress log records project activities but does not replace model configurations or prediction files needed to recreate a score.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The individual component is required to deliver reliable, timestamped QNH and dewpoint predictions three hours ahead, produce a formula-derived RH estimate when target-time temperature is available, and make its outputs available to the wider forecasting system. It must preserve consistency between training and inference features, avoid temporal leakage, and support later comparison with observed conditions.

Table 4.1. Functional requirements of the individual component

ID	Requirement	Reason / acceptance evidence
FR-01	Read historical VCBI meteorological records and preserve UTC ordering.	A reproducible source sequence must be available for modelling.
FR-02	Handle missing records and construct time-based, lagged and rolling features.	Training and inference must use a consistent feature pipeline.
FR-03	Train separate LightGBM models for QNH and dewpoint at T+3 hours.	Matches the component design reported in the research paper.
FR-04	Calculate RH from predicted T and Td using the documented equation.	Provides a transparent humidity estimate.
FR-05	Persist each output with the target timestamp and variable units.	Enables correct future-observation matching.
FR-06	Compare forecasts with later METAR observations and store verification results.	Supports monitoring of errors over time.
FR-07	Expose predictions through the backend integration layer.	Allows the wider dashboard or consuming service to access forecasts.
NFR-01	Keep data preparation and inference steps repeatable.	Supports scientific reproducibility and operational maintenance.
NFR-02	Identify stale, missing or invalid input data.	Reduces the risk of misleading or silently invalid outputs.

The source paper describes most of these functions at a system level but does not provide a complete acceptance-test report. The table is therefore a component requirements summary; individual acceptance criteria should be confirmed against the implementation and supervisor expectations.

Functional requirements in operational sequence

The component receives weather inputs, validates schema and timestamps, applies the training-compatible transformation, loads the intended model version, produces future QNH and dewpoint forecasts, derives RH from aligned temperature and dewpoint, persists the results and exposes them to the wider application. It must also support later lookup of each forecast for verification. Each stage should be tested separately rather than only testing the final dashboard.

The output contract should include component identifier, issue time, valid time, QNH and unit, dewpoint and unit, temperature provenance used for RH, derived RH and unit, model version and inference status. Verification fields may be added once an observation is matched, but prediction-time fields should not be overwritten by later verification values.

Non-functional requirements

Reliability requires clear failure behaviour when data are missing or stale. Maintainability requires shared preprocessing logic. Observability requires logs of input/output timestamps, model version and exceptions. Reproducibility requires versioned configuration and retained evaluation files. The supplied sources do not state a measured latency requirement; any such figure should only be added if it has been measured.

Security requires protected credentials, validated request payloads and no unnecessary exposure of internal database details. These are proposed deployment requirements, not claims that every control has already been implemented.

4.2 Component architecture within the complete solution

The component has a data layer, preprocessing layer, prediction layer, derived-variable calculation, persistence and verification flow, and backend interface. Historical weather records support training; live METAR observations support current inference and later verification. The QNH and dewpoint models are separate regressors. RH calculation consumes the predicted dewpoint and the predicted temperature produced by another module in the overall group solution. This makes the QNH/RH/dewpoint component dependent on a correctly aligned temperature

forecast for the same target time. This component-level architecture is illustrated in Figure 4.1.

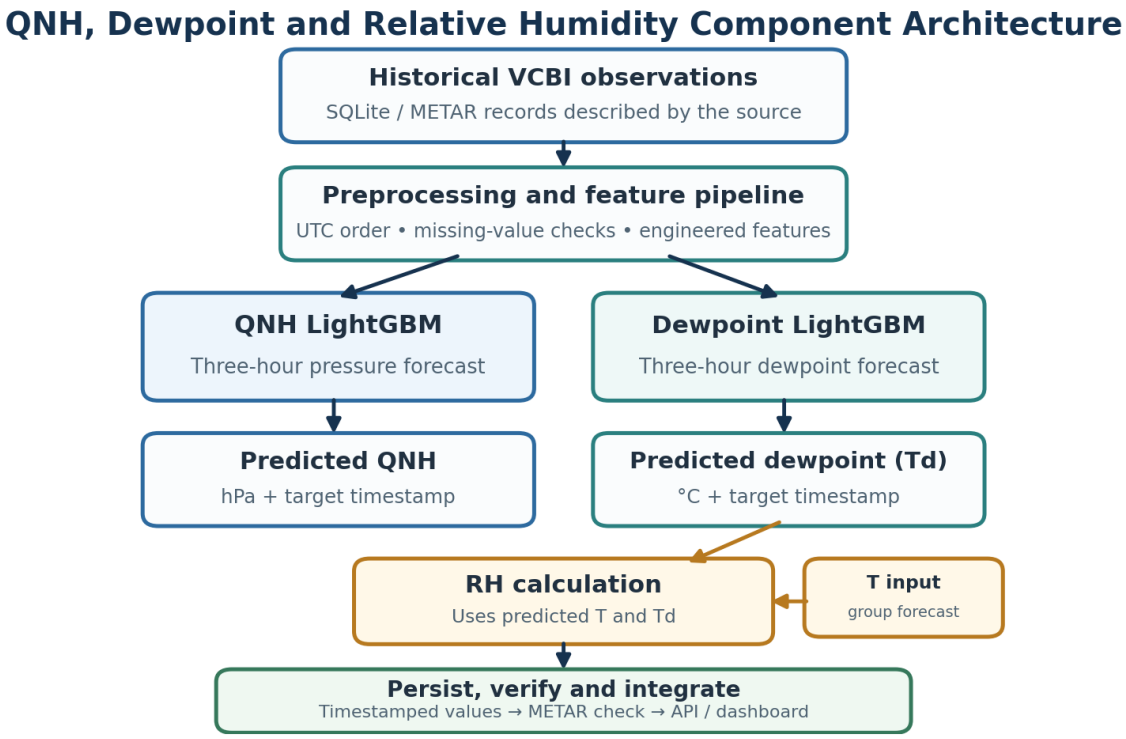


Figure 4.1. Component architecture for QNH, dewpoint and RH prediction, reconstructed from the supplied group research paper and the earlier thesis draft.

In the wider system, FastAPI is described as the backend for data retrieval, inference, model management and retraining. SQLite stores historical observations, while Supabase is used for cloud synchronisation and delivery of live information. A React-based interface displays wider system results; the source paper also identifies a Streamlit interface for cloud and visibility. Those interface details describe system integration and should not be interpreted as a separate RH-specific frontend.

A forecast record should include at least the variable, forecast issue time, target time, prediction value, unit and model version. For verification, the target time is matched to the corresponding future observation. Adding input-data status and training-window metadata would further improve auditability.

Logical layers and data ownership

The data layer provides historical records for training and current observations for inference. The modelling layer applies preprocessing and feature construction,

followed by separate LightGBM estimators for QNH and dewpoint. The physical-calculation layer derives RH after checking that temperature and dewpoint share a target time. The service layer exposes outputs and persists forecast records; the verification layer later matches forecasts to METAR observations. The presentation layer should display results but not recalculate scientific quantities independently.

This separation reduces coupling. Model replacement should not require a change to the RH equation, and a dashboard redesign should not alter the forecast target. Contracts between layers therefore need schemas and tests.

Storage and temporal consistency

A forecast record should have a unique identity or a reliable composite key involving component, model version, issue time and target time. Historical forecast records should not be overwritten when a new forecast targets the same valid time. Verification records should preserve observation source and verification time, enabling comparison across model versions.

4.3 Design alternatives and decisions

The first design decision is to forecast QNH and dewpoint separately rather than use one multi-output model. This provides an independent prediction path and makes it possible to evaluate errors in each variable's natural unit. The supplied research paper reports the use of separate LightGBM models for the QNH and dewpoint targets; it does not document a formal ablation or algorithm-comparison study for these two specific targets. Accordingly, the decision is described as the implemented architecture rather than as an experimentally proven best choice.

The second decision is to derive RH with a physical equation. An alternative would be a third model trained directly on RH records. Formula-based derivation is interpretable and enforces a known relationship between temperature and dewpoint, but its accuracy depends on the quality of the two predictions and the suitability of the approximation. A future comparison should evaluate equation-derived RH against an independent RH model using the same chronological holdout.

The third decision concerns the forecast horizon. The individual QNH/dewpoint workflow targets T+3 hours, which aligns with the principal horizon stated in the

group paper. Changing the horizon would require creating new shifted targets and re-evaluating each model; results at a different horizon should not be inferred from the reported T+3 scores.

Why separate pressure and dewpoint estimators

Separate estimators allow each target to have its own learning objective and evaluation. Pressure and dewpoint have different units and dynamics, and separate models make it easier to diagnose which target is weak. The progress log records early multi-output backend support, while the final paper describes separate QNH and dewpoint models. The final thesis should represent the final implementation as separate models unless the repository proves otherwise.

Why derive RH using a physical equation

The design options are a direct RH model or a calculation from temperature and dewpoint. The paper describes the second design using the August-Roche-Magnus relationship. This makes the dependency explicit, but RH inherits errors in the upstream forecasts. Any alternative RH model would need a controlled comparison over the same timestamps.

Choice of LightGBM and baseline discipline

The paper identifies LightGBM as the final model family for the component, and the progress log records earlier exploration of Random Forest and XGBoost. That documents a model-selection process but does not prove global superiority. A final benchmark needs the alternative models evaluated on identical test cases with comparable settings.

4.4 Detailed design

The preprocessing stage starts by ordering the source records chronologically in UTC. The supplied paper states that missing values are filled but does not specify whether this uses forward filling, interpolation, a constant value or a variable-specific rule. That detail is material: a chosen imputation method should not use future target information and should be documented in the final experiment record.

The feature set described for the QNH/dewpoint subsystem includes hour-of-day and time-of-year features in sine and cosine form, the difference between temperature and dewpoint, QNH movement, and three-hour and six-hour rolling averages. These features represent periodic behaviour, recent moisture/temperature relationships and short-term pressure trends. Rolling statistics must be computed only from records available at the prediction issue time. Any feature derived from the future target period would cause temporal leakage.

For training, the target series is shifted forward three hours and the records are split chronologically into 80% training and 20% testing, without shuffling. Separate LightGBM models are then fitted for QNH and dewpoint. During inference, the same feature names, order, units and preprocessing rules must be applied. The paper reports automated retraining as part of the system lifecycle, but it does not specify the exact cadence and deployment details solely for the component beyond general periodic/monthly retraining statements.

$$\mathbf{RH\ (\%) = 100 \times \exp[(17.67\ Td / (Td + 243.5)) - (17.67\ T / (T + 243.5))]}$$

Here, Td is predicted dewpoint in °C and T is predicted dry-bulb temperature in °C. Values must be aligned to the same target time. If T or Td is absent, stale or outside the expected numeric domain, the RH output should be marked unavailable rather than silently computed from a mismatched prediction.

RH calculation inputs and safeguards

The equation uses predicted dewpoint Td and dry-bulb temperature T, both in degrees Celsius. Before calculation, the service should check that values are numeric and finite, units are correct and target timestamps match. The result should be checked for finiteness and plausibility. A value outside the expected physical range should be flagged according to a documented policy rather than silently clipped without a record.

The formula should live in one shared function and have deterministic tests. This reduces the chance that backend and dashboard code implement slightly different equations. Tests should cover ordinary inputs, temperature close to dewpoint, missing fields, invalid strings and timestamps that do not match.

Feature pipeline design

The progress log states that feature-selection logic was encapsulated within preprocessing pipelines. This helps inference receive the expected feature vector rather than relying on individual callers to recreate transformations. Each model should have a versioned list of feature names, types, time offsets, units and missing-value rules. Required missing features should be handled explicitly; they should not be filled with arbitrary zeros unless that choice has a justified meaning.

4.5 Standards and professional practice

The system uses METAR and TAF terminology and is positioned as an aviation-weather decision-support prototype. The International Civil Aviation Organization's meteorological service material and the Federal Aviation Administration's aeronautical knowledge guidance are included in the supplied paper's references [14], [15]. These sources are relevant to the context in which pressure and wind-related quantities are interpreted, but the supplied materials do not demonstrate a formal compliance assessment or certification of this software component.

Professional practice requires recording the unit and timestamp of every value, keeping observation and prediction separate, documenting assumptions and limitations, and distinguishing model estimates from authoritative meteorological reports. When values are surfaced to users, the interface should include data freshness and a reminder that model output is supplementary. The thesis must not claim operational approval, regulatory certification or a demonstrated safety benefit unless supported by separate evidence.

Documentation and review discipline

Professional practice includes traceable requirements, consistent units, explicit model limitations, source acknowledgement, reviewable tests and repeatable releases. Every reported number should trace to the group paper or an actual output file. Every figure should have a caption, and tables should distinguish observed results from design proposals and test plans.

The component should be described as research and decision-support software unless formal certification or acceptance is documented. An API and dashboard do not

themselves establish compliance with aviation service standards. Any future production use requires applicable organisational and regulatory review.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The group paper identifies Python-based machine-learning implementation, LightGBM for QNH and dewpoint, SQLite/SQLite3 for local historical data, Supabase for cloud data synchronisation, and FastAPI for backend services. The complete system uses a React.js dashboard for general weather, prediction and warning displays, while Streamlit is identified for the cloud and visibility module. The individual component therefore needs to remain compatible with the shared data and API contracts rather than introduce a separate stand-alone dashboard.

The provided paper does not specify Python, LightGBM, FastAPI, database or frontend version numbers. It also does not name all preprocessing libraries or show a dependency lock file. Those implementation details should be taken from the project repository rather than guessed. Version control, pinned dependencies and configuration outside source code are recommended so that a later retraining run can be compared with the one that generated the reported metrics.

Environment and dependency record

The paper identifies Python-based ML services, LightGBM for QNH/dewpoint, FastAPI for backend services, SQLite for local historical data, a cloud database service for parts of the wider system and a React-based dashboard. The progress log refers to Pipeline.py, API work, training/verification scripts and automated retraining. The exact Python and package versions, host, lockfile and schema are not supplied here and should be copied from the actual project configuration.

The environment manifest should distinguish runtime dependencies from development dependencies. Pinning package versions where possible helps future reviewers load saved estimators and reproduce the inference pipeline.

5.2 Component implementation

The data preparation implementation reads the historical weather series, sorts records by UTC and creates predictors that encode time cycles, recent differences and rolling statistics. The target values are shifted three hours forward. The chronological split

creates a training segment followed by a later test segment, so the reported test period is not randomly mixed with the training observations. Exact transformation code and missing-data rules should be referenced from the repository in the final version.

The QNH model and dewpoint model are trained independently using LightGBM. QNH predictions are evaluated in hPa and dewpoint predictions in degrees Celsius. The RH routine receives predicted dewpoint and predicted dry-bulb temperature and evaluates the stated exponential relationship. The RH routine is deterministic given the two input values, so model performance and formula implementation should be tested separately: model tests check input forecast accuracy, while calculation tests check units, numeric boundaries and equation output.

The operational pipeline stores forecasts with their target timestamps and later uses METAR observations to identify the matching target record. This allows error calculations to be made after the target observation arrives. Since the source paper states that the wider system supports background data acquisition and model retraining, the individual component should expose clear input/output boundaries and avoid embedding database or API credentials in model code.

Initial model exploration and final model path

May entries record a Random Forest QNH regressor, backend support for multi-output predictions, API/frontend structure and training/verification scripts involving LightGBM and XGBoost. The log later records preprocessing improvements and removal of obsolete model scripts. This documents iteration. The final thesis should not describe all historical scripts as active; the final path must be confirmed against the repository.

Three-hour LightGBM forecasting

The August 18 log records implementation of LightGBM three-hour models. The paper states that QNH and dewpoint targets are shifted three hours forward and data are split chronologically. The implementation therefore needs an explicit target shift and a saved model paired with the preprocessing definition. Prediction files should

retain target timestamps and sample counts. Rows lost through lag features or missing values should be counted separately from raw records.

RH formula implementation

The August 21 log records the August-Roche-Magnus formula being implemented to derive RH from predicted dewpoint and temperature. It is a calculation stage after inference, not a third independently trained model. Code should record the temperature source and ensure it shares the target time with dewpoint. Formula tests should be run separately from data-acquisition and API tests; if no test logs exist, they should be described as required work rather than completed tests. The calculation sequence is illustrated in Figure 5.1.

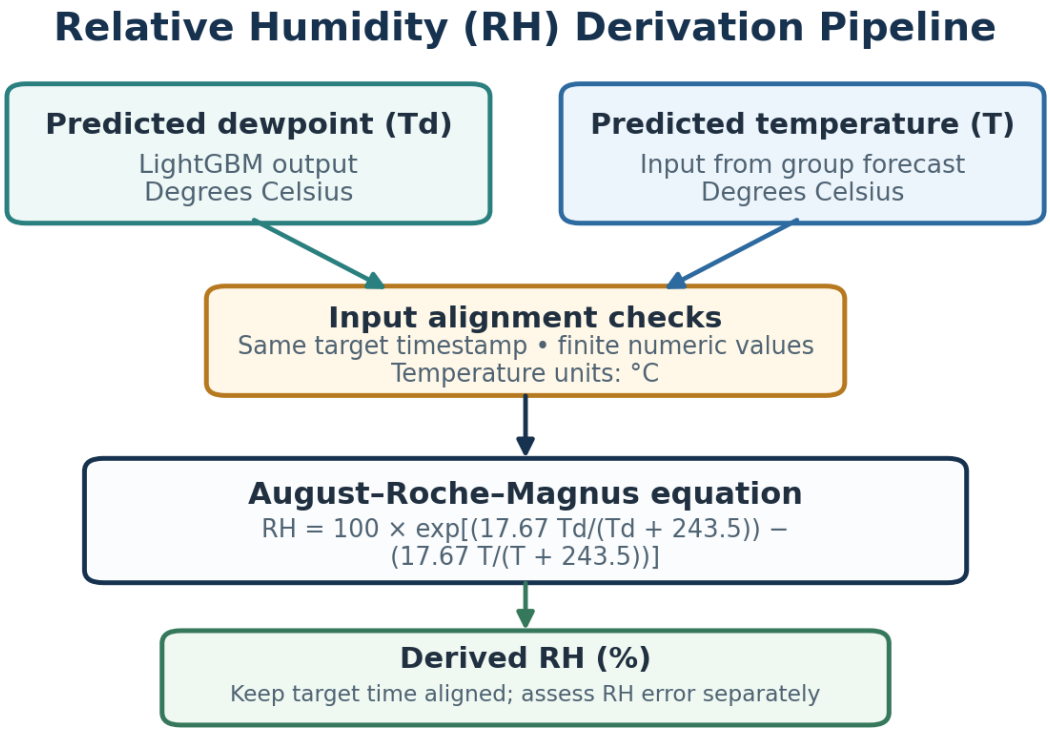


Figure 5.1. RH is derived from predicted dewpoint and the temperature forecast supplied by the wider group system, rather than from a separate RH model.

Feature selection and preprocessing maintenance

The log says feature selection was encapsulated within preprocessing pipelines. Training and inference should use the same feature-generation version. Feature-

importance charts, where retained, are useful for exploratory analysis but should not be interpreted as causal evidence about atmospheric processes.

5.3 Integration with the other components

The RH calculation has a direct dependency on the group temperature forecasting subsystem, while QNH and dewpoint are forecast by the individual component. The integration contract must guarantee that the supplied temperature value is a prediction for the same target time as the dewpoint value, not the latest observed temperature or an output from a different forecast cycle. The QNH output can be integrated independently, but should carry the same forecast issue/target-time metadata.

At the system level, FastAPI supports data retrieval, inference, verification and model-management functions. Supabase supports synchronisation and broadcasting of live information, while SQLite retains local observations and associated records. An integrated request should return clear status information if a needed input is missing or the selected model is not available. The source paper describes the integration architecture but does not report API response times, end-to-end latency, or an independently measured service-availability percentage [16].

Integration dependencies and version alignment

The paper describes a shared service environment for retrieval, inference, training and verification. The progress log records API/database planning, alignment with the group master branch, removal of wind-prediction features from this individual scope and placeholders for other members' models. This demonstrates interface and scope-management work.

RH depends on the temperature forecast from the wider system. The interface should establish whether temperature is provided directly or retrieved by a forecast identifier. If the upstream component is unavailable, RH should be flagged unavailable. The newest observed temperature should not silently replace a missing future forecast for a three-hour target.

Database consistency and forecast versioning

If SQLite and a cloud database both participate in the wider system, their ownership and synchronisation path should be explicit. Each inference should preserve an immutable forecast record. Retraining may update the active model pointer but should not erase the model version associated with earlier forecasts.

5.4 Verification and software testing

The main quantitative verification evidence available in the supplied paper is the QNH MAE and dewpoint MAE/R^2. The broader framework saves each forecast with a target timestamp and compares it with the future METAR observation when available. This is necessary to ensure that each prediction is evaluated at the target horizon rather than compared with the issue-time observation.

The paper does not provide a formal unit-test report, integration-test coverage, test case IDs, or an RH-specific validation score. It would be inaccurate to claim that these tests passed based solely on the reported modelling metrics. Appendix A therefore includes a proposed verification checklist that can be completed against the actual code and run logs before final submission.

At minimum, testing should cover the presence and type of required features, consistent handling of missing inputs, correct three-hour target alignment, deterministic RH calculation for fixed T/Td values, rejection of unaligned or stale temperature predictions, database persistence, and successful comparison of a forecast with the later METAR record. Tests should also exercise monsoon-period slices and missing-data cases if the necessary records are available.

Test layers

Unit tests can validate the RH equation, feature construction and input validation. Pipeline tests can check that training and inference create the same feature names and shapes. Model-load tests should load saved estimators and predict from a valid row. API tests should check schemas and failure responses. Integration tests should verify database persistence and forecast/METAR matching. Regression tests can compare a deterministic known input with an expected output.

The supplied materials do not include a complete executed software-test report. Test matrices in this draft are proposed plans, not evidence that the tests passed. Mark a case passed only after running it and retaining output.

Verification matching tests

Verification tests should cover exact matches, missing observations, duplicate observations, timestamp tolerance and delayed METAR arrival. A repeated verification job should not create duplicate records. Forecasts with no observation should remain marked unverified rather than being assigned zero error.

5.5 Deployment and operation

The group system is described as retrieving live METAR data, parsing it, storing it and generating forecast outputs for a dashboard. Background tasks support data acquisition, database synchronisation and periodic retraining. The component should use the same operational process, with the model version and training cutoff retained so that the forecast can be traced to the data available at inference time.

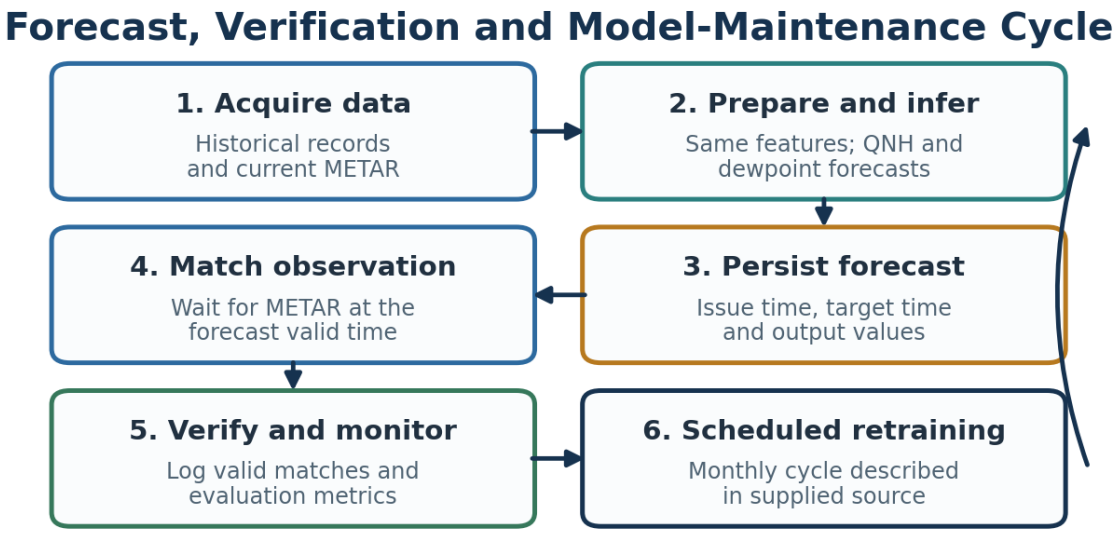
Monthly retraining is mentioned in the wider paper as a way to incorporate new observations and respond to changing weather behaviour. This should be treated as the described system design, not as proof that every scheduled run completed successfully or improved the model. A practical deployment log should record the start/end time, data window, model artefact identifier, validation score and whether the newly trained model passed acceptance criteria before replacing the active model.

Continuous training and model lifecycle

The paper describes automated monthly training and live validation, and the August progress log records a dedicated auto-retraining utility. A controlled flow should define a training window, validate data quality, fit a candidate, run checks, compare it with the current model and promote only if acceptance rules pass. Failed runs should leave the active model unchanged.

Retraining is not automatically beneficial: incomplete or anomalous data can degrade a model. The lifecycle should record data window, feature pipeline, configuration, metrics and promotion decision. Rollback should be possible if the new model performs poorly. The sources confirm retraining automation but do not fully

document a guarded promotion/rollback policy. Figure 5.2 summarises the forecast, verification and model-maintenance cycle.



Promotion of updated weights should use a documented chronological validation gate.

Figure 5.2. Operational sequence for forecast persistence, future METAR verification and scheduled model maintenance.

Operational monitoring

Monitoring should track data freshness, successful forecast counts, missing inputs, inference failures, verification coverage, model version and error metrics when observations arrive. System health must be separated from forecast skill: a healthy API can return poor forecasts, while a strong offline model can be inaccessible due to a service fault.

6 RESULTS AND VALIDATION

6.1 Research results

The supplied group research paper reports the following outcomes for the QNH/dewpoint component. Dewpoint prediction with LightGBM achieved an R^2 of approximately 0.66 and an MAE of approximately 0.58 °C. QNH prediction achieved an MAE of approximately 1.8 hPa. These are the only component-specific numerical results reported in the materials supplied for this draft [16].

Table 6.1. Reported QNH and dewpoint evaluation results

Output / method	Reported result and evidence limit
Dewpoint (T+3 h), LightGBM	MAE ≈ 0.58 °C; R^2 ≈ 0.66. The test sample count, full residual distribution and exact split boundaries were not supplied.
QNH (T+3 h), LightGBM	MAE ≈ 1.8 hPa. QNH R^2 and RMSE are not reported in the source paper.
Relative humidity, August–Roche–Magnus using predicted T and Td	No separate RH accuracy metric is reported. Direct validation is required before an RH error claim can be made.

The reported metrics should be understood as approximate summaries from the group paper. The raw held-out prediction file, sample count, confidence interval and split boundaries were not attached, so this thesis cannot independently recompute them. The results are reproduced faithfully rather than expanded with invented decimal places or unreported metrics.

Component-level result table and interpretation

The following values are transcribed from the supplied group research paper and were not newly computed by this thesis. Dewpoint has a reported R^2 of approximately 0.66 and MAE of approximately 0.58 °C. QNH has a reported MAE of approximately 1.8 hPa. The paper does not supply a separate numerical RH error metric. Since prediction arrays, exact test count and holdout dates were not supplied, these values remain reported summary results pending reproduction from experiment outputs.

The dewpoint metrics indicate fit relative to observed variation in the evaluation set but leave room for error. MAE does not describe maximum error or seasonal bias. QNH MAE provides a pressure-unit summary, but interpretation requires the test window, pressure-change distribution and baseline results. Residual plots and

seasonal breakdowns should be added if raw predictions are available. The reported numerical results and the RH evidence gap are summarised in Figure 6.1.

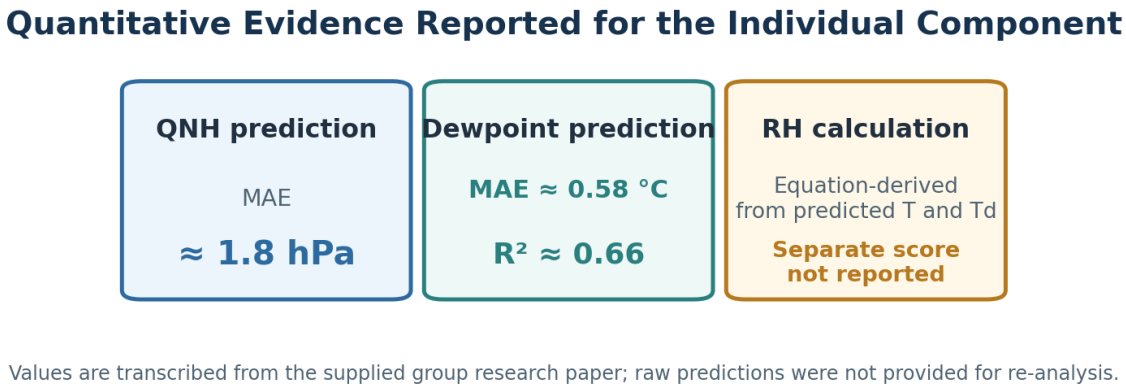


Figure 6.1. Summary of metrics reported for QNH and dewpoint; a separate RH validation score was not provided.

RH result boundary

The RH formula is described in the paper and recorded as implemented in the progress log. This supports the method description, not the predictive accuracy of RH on held-out cases. A direct RH evaluation needs predicted temperature and dewpoint for the same test timestamps, a consistent reference and actual MAE/RMSE/bias calculations.

6.2 Product and technical results

The component delivers three related outputs: a QNH forecast, a dewpoint forecast and an equation-derived RH estimate. The QNH and dewpoint models generate separate numerical predictions three hours ahead; the RH calculation combines predicted dewpoint with the wider system's predicted dry-bulb temperature. Keeping the outputs separate makes it possible to trace the RH value to the two meteorological predictions on which it depends.

The wider implementation described in the group paper has a live METAR acquisition path, persistent observation and prediction records, FastAPI backend services, and dashboard integration. The system is intended to support ongoing verification and retraining. No independent technical measures such as service

latency, number of successful scheduled jobs, endpoint uptime, user acceptance scores or RH calculation throughput are provided in the source. These should be added only if actual logs or test results exist.

Technical output versus research outcome

The technical outcome is a component that exposes QNH/dewpoint prediction capability, derives RH and is designed to integrate with the forecast service. The research outcome is evidence about correspondence with later observations. A working pipeline or dashboard demonstrates implementation progress but not generalisable forecast skill.

The progress log documents the QNH Random Forest prototype, training and verification scripts, LightGBM finalisation, formula implementation, MLOps utility and integration tasks. The final submission should attach real screenshots or repository evidence only where they exist and are approved for inclusion.

6.3 User or stakeholder validation

The supplied research paper acknowledges guidance from meteorological professionals, but it does not report a formal usability study, structured stakeholder questionnaire, acceptance-test sign-off or operational trial for the QNH/RH/dewpoint component. Acknowledgement of expert guidance should not be treated as equivalent to measured user validation.

For a complete validation record, the project could ask a qualified meteorology stakeholder to review forecast plots, error summaries, units, freshness indicators and the RH derivation. Feedback should be recorded with the participant role, date, review criteria and resulting changes, subject to institutional and data-governance requirements. Until evidence is available, the present thesis states that formal stakeholder validation has not been documented in the materials supplied.

Status of stakeholder validation

The sources mention meetings with a meteorology department and an aviation supervisor, but do not provide a formal user study, participant count, survey, interview transcript, acceptance score or structured usability evaluation. This thesis therefore does not report stakeholder-satisfaction numbers. The meetings are

consultation events in the log; they should not be described as formal validation unless meeting notes or a signed evaluation form substantiate that claim.

A future review could ask whether target times, units, freshness indicators, uncertainty labels and verification status are understandable. Any required participant study should use an approved instrument and follow university rules. Suggested questions are not evidence of a completed human-participant study.

6.4 Integration results

The research paper describes how the individual outputs fit into a larger architecture that handles current METAR observations, prediction services, storage, validation and frontend presentation. It also describes the RH calculation as part of a broader collection of forecasts and aviation parameters. This supports a qualitative conclusion that the component was designed for integration with the shared system.

The source does not include an interface contract, response examples, integration-test logs or measurements of end-to-end latency. Therefore, this draft does not assert specific API performance, successful transaction counts or a guaranteed level of availability. The final implementation chapter should include actual request/response examples and screenshots or test logs if the repository provides them.

Integration evidence and remaining checks

The sources support that backend API work, database integration, model inference and automated workflow components were developed. The progress log notes integration placeholders and branch alignment. A complete integration result should include the tested release, endpoint behaviour, database examples and proof that forecast and observed timestamps are correctly paired.

A key risk is mismatched provenance. If QNH and dewpoint forecasts are from one issue time but temperature comes from another, derived RH is not a consistent forecast vector. A final end-to-end trace should show issue time, target time, model versions, temperature/dewpoint inputs, derived RH and later verification observation.

6.5 Summary of findings

The available evidence supports two quantitative findings: the dewpoint model has a reported MAE of about 0.58 °C and R^2 of about 0.66, while the QNH model has a reported MAE of about 1.8 hPa. The RH component is implemented as a physical calculation from predicted temperature and dewpoint, but an independent RH accuracy score is not reported. The wider architecture supports time-stamped storage and future-observation comparison, although operational and stakeholder-validation metrics are not present in the supplied paper.

These findings are sufficient to describe an implemented forecasting component, but they are not sufficient to establish uniform performance across every monsoon season or to claim readiness for safety-critical operational use. Further evidence should focus on seasonal holdouts, baseline comparison, RH-specific evaluation and clear reproducibility artefacts.

Evidence matrix

The available evidence supports the development sequence in the progress log, the numerical QNH/dewpoint summaries in the paper, and the RH calculation pathway. Exact RH accuracy, full prediction diagnostics, seasonal metrics and formal stakeholder evaluation are unreported in the supplied materials.

The evidence-bounded conclusion is that the component demonstrates a defined modelling and calculation workflow, with initial numerical evidence for QNH and dewpoint. Stronger operational claims require reproducible experiment artefacts, separate RH validation, comparisons with simple baselines and evaluation across representative monsoon periods.

7 DISCUSSION

7.1 Interpretation of findings

The dewpoint MAE of approximately 0.58 °C indicates that the reported absolute prediction error is below one degree Celsius on average for the evaluated sample, while R^2 of approximately 0.66 indicates a meaningful but incomplete fit to target variation under the standard coefficient-of-determination definition. These figures suggest useful predictive signal, but without the test sample count, variance, residual plots and seasonal breakdown it is not possible to determine how errors behave during rapid weather transitions or rare conditions.

The QNH MAE of approximately 1.8 hPa provides a measure of average absolute pressure error. Because no QNH RMSE, R^2, baseline error or operational tolerance analysis is reported for this component, the metric should not be used alone to claim superiority or suitability for a specific operational procedure. A comparison against persistence and a meteorologically appropriate baseline would place the number in context.

RH is mathematically linked to temperature and dewpoint. An accurate dewpoint prediction does not automatically guarantee an equally accurate RH prediction because the temperature estimate also contributes to the calculation. The derivative of the equation varies across the temperature range, so the same input error can have different RH effects in different atmospheric conditions. Direct RH verification is needed, especially for periods close to saturation where RH changes can be operationally meaningful.

Interpretation under changing monsoon conditions

Weather behaviour and input distributions can shift across seasonal regimes. A model may learn recurring patterns but remain less reliable during transitions or rare events. Aggregate metrics cannot show whether errors are evenly distributed among the four Sri Lankan seasonal regimes. The monsoon framing should therefore be followed by season-aware validation rather than a claim of equal performance everywhere.

A future analysis should group residuals by month and by approved definitions of Southwest Monsoon, Northeast Monsoon, First Inter-Monsoon and Second Inter-Monsoon. Report sample counts with every subgroup and distinguish model errors from data gaps or delayed observations.

Interpretation of model-specific metrics

Dewpoint MAE near 0.58 °C gives an average absolute error; R^2 near 0.66 describes variation captured relative to a mean baseline within the evaluation sample. QNH MAE near 1.8 hPa is a pressure-specific summary. None alone indicates calibration, signed bias, temporal clustering of large errors or performance against persistence.

7.2 Comparison with literature baselines and existing solutions

The literature in the group paper establishes relevant context for applying machine learning to general weather forecasting and aviation meteorology [1], [2], [4], [6]–[10]. However, the reported results for QNH and dewpoint are not compared in the source paper with an external model evaluated on the same VCBI records, feature set, forecast horizon and test interval. A fair benchmark would require identical chronological splits and target timestamps.

The broader research paper reports comparisons for other components, such as wind and cloud/visibility prediction, but those numbers do not constitute QNH or dewpoint baselines and should not be assigned to this individual component. In future work, this component could compare LightGBM with persistence, linear regression, Random Forest and another gradient-boosting implementation. The comparison should report MAE and RMSE for both targets, R^2 where appropriate, and season-level results.

Limits of cross-study comparison

The cited studies differ in target variables, spatial scales, inputs, methods and evaluation designs. Rainfall prediction [1] and general short-term weather prediction [2] provide context but are not directly comparable QNH/dewpoint baselines. Aviation weather studies motivate local information, while ML papers show that nonlinear time-dependent approaches are relevant. Numeric superiority must not be claimed without aligned tasks and metrics.

Within-project comparisons would be more informative. Random Forest and XGBoost were examined during development, but the supplied materials do not include their final numerical results on the same QNH/dewpoint test set. Add comparisons from saved logs if available; otherwise report this as a limitation.

7.3 Research contribution

The research contribution is a focused airport-level workflow for predicting two meteorological variables three hours ahead and deriving a third related variable with a transparent physical equation. The component combines historical time-series preparation, separate LightGBM regressors, a thermodynamic relationship for RH and a future-observation verification process within a larger aviation-weather system.

Its contribution is therefore both methodological and applied: it makes the forecast pipeline explicit, keeps QNH and dewpoint errors in interpretable units, and provides a route to inspect the derived RH output. The evidence does not yet establish generalisation to other airports or all monsoon regimes; those are future evaluation questions rather than proven contributions.

7.4 Product contribution

At product level, the component adds pressure, dewpoint and humidity information to the wider airport-weather interface and verification record. Forecasts can be stored with target timestamps, made accessible through backend services and compared with observed values when future METAR reports arrive. This can support monitoring and analysis by users who need short-horizon local weather context.

The product value remains a potential capability rather than a measured operational outcome. The materials supplied do not include a formal stakeholder acceptance study, documented reduction in weather-related disruption, or evidence that the output has been adopted in operational flight decisions. Such claims should only be made after a governed pilot and measured evaluation.

7.5 Limitations and threats to validity

Several limitations affect interpretation. First, the source reports aggregate metrics without the exact number of test samples, prediction window, confidence intervals or

full residual distribution. Second, the reported dataset covers 2019–2026, but the paper does not provide season-wise or year-wise scores. Third, the missing-value strategy is not detailed. Fourth, the exact feature list and LightGBM parameters are not included. Fifth, an independent RH error metric is absent.

Temporal leakage is another potential threat. Rolling features and normalization, if applied, must be constructed using only information available at the forecast issue time, and any fitted transformation must be fit on the training subset before being applied to the test subset. The paper describes chronological splitting and no shuffling, which helps, but the exact implementation is not included for independent audit.

External validity is limited because the study is focused on one airport. Model behaviour may change across Sri Lankan monsoon seasons, rare weather events, data-provider changes or other airports. The reported model metrics also do not in themselves demonstrate compliance with aviation operational standards. Finally, the supplied paper contains reference-number inconsistencies that need correction before the thesis bibliography is finalised.

Threats related to internal validity

Internal-validity risks include target-time misalignment, leakage, duplicated records, unsuitable imputation, inconsistent units, differences between training and live preprocessing, and incorrect joins between forecasts and METAR observations. A chronological split mitigates one risk but does not automatically remove all leaks or alignment errors.

Threats related to external validity

The system is described for VCBI, and reported metrics do not show season-wise performance, unusual weather episodes or results on future years. Results should not be generalised to other airports or every monsoon period without local validation. The dataset time span alone does not prove equal coverage of each regime.

Measurement and reporting limitations

The supplied sources omit exact test counts, split dates, prediction arrays, full parameter configurations, a standalone RH error, residual plots and formal user

evaluation. These omissions limit independent reproduction; they do not prove that the model is inaccurate, but they bound the strength of the evidence.

7.6 Implications

For research, the component illustrates a practical way to combine supervised learning with a physical relationship for a related meteorological quantity. It also shows why verification should be built into the system rather than left as a one-time test. Keeping predictions with their target timestamps enables model performance to be monitored as new observations accumulate.

For potential operational users, the implication is cautious: QNH and dewpoint forecasts may add context to the current observation stream, and derived RH may be informative when its inputs are aligned and reliable. The outputs should remain clearly labelled as model predictions and should be reviewed alongside official products and qualified judgement. Before stronger claims are justified, the system requires season-specific validation, tested failure handling, a reproducibility record and direct RH evaluation.

Implications for model governance

An operational prototype should apply release acceptance checks for data quality, timestamps, headline metrics and changes relative to the currently deployed model. If error grows or verification samples are too small, the system should flag the condition instead of assuming retraining improved performance.

A derived RH value should carry the identity of the temperature and dewpoint forecasts used. This allows later explanation and diagnosis. The same principle applies to systems in which data-driven predictions feed a physical calculation.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The potential value proposition is a local, short-horizon forecast of QNH and dewpoint together with a derived RH estimate that can be inspected alongside current METAR observations. Candidate users include airport meteorological staff, aviation operations analysts, researchers, and developers evaluating weather-based decision-support tools. These are prospective user groups, not confirmed customers; the provided paper does not include a market survey or customer interviews.

The benefit offered is the consolidation of recent data, a forecast target time, a small set of interpretable output variables and an eventual verification record. To establish practical value, a pilot would need to determine whether the predictions add information beyond the existing observation/forecast products and whether the interface and latency fit actual user workflows.

Value proposition boundaries

The immediate value proposition is an airport-local source of forecast context with a path to later verification, not a replacement for official meteorological products. Potential users include project reviewers, meteorological analysts, airport-operations stakeholders and developers integrating weather information. The supplied files do not establish external adoption or purchasing.

Useful value measures include forecast availability, distinction between forecast and observation, error-detection speed, visible data gaps and improvement over a simple baseline. The thesis describes potential value rather than measured cost savings or safety outcomes.

8.2 Market and competitive position

The component sits conceptually alongside existing meteorological services, airport weather displays and general-purpose forecasting tools. Its possible distinction is focus on one Sri Lankan airport, short forecast horizons, explicit RH derivation and comparison against future METAR observations. The supplied materials do not

document a commercial competitor analysis, prices, licensing comparisons or accuracy evaluations against commercial products.

A defensible competitive assessment would need to identify products authorised for the target operational context, confirm allowed data and API use, compare forecast horizons and target variables, and use a common evaluation period. Until this work is performed, the project should be described as a research prototype with a potential local focus, not as a commercially superior or certified aviation service.

Positioning against existing services

The component can be positioned as a locally tailored research prototype combining prediction and verification. Existing aviation-weather services may differ in authority, scope, update frequency and validation standards. The supplied materials do not contain a market survey, competitor-price comparison or customer interviews, so they do not support a quantified competitive-advantage claim.

8.3 Business or sustainability model

A possible sustainability approach is to maintain the software as a research or institutional decision-support service, with scheduled data ingestion, model monitoring, retraining and periodic technical review. Costs would likely include service hosting, database storage, monitoring, engineering time and validation activity; actual amounts have not been supplied and should not be estimated without deployment requirements.

If the system were evaluated for broader use, a staged model could begin with a controlled research pilot, followed by a documented stakeholder evaluation and a governed service arrangement. This would need clear responsibility for data quality, model updates, incident handling and withdrawal of stale or degraded predictions. A business model should not be finalised before operating costs, data rights and user requirements are understood.

Operating cost categories to measure

A sustainability plan should account for data access and storage, inference and retraining compute, application hosting, monitoring, maintenance and staff time for investigating anomalies. These are cost categories, not measured costs for this

project. A later pilot could track monthly infrastructure use and maintenance hours before proposing a service or pricing model.

The project could remain an academic prototype, become an internal decision-support module or be integrated into a wider service. Each route carries different uptime, support, data licensing, access-control and release-management obligations. The supplied materials do not establish which route has been chosen.

8.4 Scalability, intellectual property and compliance

The modular structure offers a potential path to scaling the service, but moving from VCBI to another airport would require data availability checks, local feature validation, model retraining and fresh evaluation. The current reported results should not be assumed to transfer directly to another site. The same applies to changes in data-provider schemas or observation frequency.

Intellectual-property ownership, licences for third-party libraries, database terms and rights to redistribute meteorological data should be reviewed before any public release. The supplied materials do not establish a legal ownership or licensing position. Aviation standards and formal regulatory obligations should also be assessed by appropriate professionals; inclusion of ICAO and FAA material in the reference list is not itself proof of compliance [14], [15].

A scalable implementation should add model registry records, health checks, automated data-quality alerts, clear rollback mechanisms and a record of retraining outcomes. These are proposed hardening measures. The thesis should distinguish them from features already confirmed by the project repository.

8.5 Contribution to the Sustainable Development Goals

The project has a potential indirect relationship to Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) through applied digital and predictive infrastructure, and to Goal 13 (Climate Action) through the use of data to understand weather variability. This relationship is a proposed alignment, not a measured SDG outcome. The present study does not quantify emissions reductions, improved resilience, or changes in airport performance.

A responsible sustainability claim should be supported by evidence of who uses the system, what decision it improves, how reliable it is across the relevant conditions and whether the service can be maintained without creating unacceptable reliance on weak forecasts. Future evaluation could define metrics for data freshness, model performance over time, user-reported usefulness and maintenance effort before claiming a realised sustainability benefit.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This individual component addresses short-horizon prediction of QNH and dewpoint for VCBI and derives relative humidity from the predicted dewpoint and predicted dry-bulb temperature. The design combines time-aware preprocessing, separate LightGBM regressors, a physically interpretable equation and a verification workflow in which forecasts are matched to future METAR observations. These functions support the larger hyper-local aviation weather forecasting and verification system described in the group research paper.

The reported results are a dewpoint MAE of approximately 0.58 °C with R^2 of approximately 0.66, and a QNH MAE of approximately 1.8 hPa. The evidence for RH is currently the implemented calculation method rather than a separate accuracy result. The available numbers suggest the component learns useful signal from historical records, but the evidence is not enough to claim uniform performance across every monsoon season or operational readiness for safety-critical use.

Final synthesis of evidence

The individual component contributes two three-hour machine-learning estimates and one physically derived humidity value to the wider VCBI aviation-weather system. The paper reports LightGBM for QNH and dewpoint, chronological data preparation and the August-Roche-Magnus relationship for RH. The progress log provides a development trail from early QNH modelling and pipeline work through final LightGBM models, RH calculation and MLOps tasks.

The quantitative conclusion is specific: dewpoint MAE about 0.58 °C and R^2 about 0.66, and QNH MAE about 1.8 hPa according to the paper. A separate RH error score is not reported. To strengthen the evidence, reproduce metrics from saved predictions, validate RH directly, publish exact data windows and sample counts, compare baselines and examine seasonal errors.

9.2 Achievement of objectives

Table 9.1. Objective achievement and evidence summary

Objective	Evidence and status based on the supplied paper
Prepare historical VCBI data and temporal features	Dataset period, sample scale, UTC ordering and feature families are described; exact imputation and split records need confirmation.
Train QNH and dewpoint models	Separate LightGBM models and T+3 horizon are described; full hyperparameters were not supplied.
Derive RH from predicted T and Td	The August–Roche–Magnus equation is specified; an independent RH score is not reported.
Evaluate model performance	QNH MAE and dewpoint MAE/R^2 are reported; raw predictions and holdout size were not supplied.
Integrate with backend and verification	FastAPI, SQLite/Supabase and future METAR matching are described; formal integration test logs were not supplied.

The objective review uses only the evidence currently available in the research paper. Where code, configuration or run logs are needed to confirm an achievement, the status should be updated after those artefacts are inspected.

Objective achievement status for review

The evidence supports that preprocessing and feature generation were addressed, LightGBM QNH/dewpoint models were implemented, the RH equation was implemented and integration/retraining utilities were developed. Exact preprocessing parameters and feature schema still need attaching. Evaluation is partially evidenced by reported metrics, while direct RH evaluation and complete run metadata remain open. Deployment acceptance and full production monitoring evidence are not included in the supplied files.

9.3 Individual research and product contributions

The individual research contribution is the QNH, dewpoint and RH prediction workflow. The individual development contribution, as stated by the student, includes preprocessing, LightGBM training, formula implementation, evaluation and backend development. The product contribution is a set of time-stamped forecast outputs designed to integrate into the wider system’s storage, API and verification process.

The key distinction is that QNH and dewpoint are machine-learning forecasts, while RH is derived from predicted temperature and dewpoint. This distinction supports

interpretability and makes it clear which outputs have independently reported numerical evaluation and which require further validation.

9.4 Future directions

1. Report the exact chronological training and test windows, number of evaluated observations, missing-value strategy, feature schema, model parameters and software versions.
2. Evaluate performance separately for the Southwest Monsoon, Northeast Monsoon, First Inter-Monsoon and Second Inter-Monsoon, and inspect error behaviour during rapid pressure and humidity changes.
3. Compute RH from predicted temperature and dewpoint for the holdout period and report MAE, RMSE, bias and error distribution against target-time observed or consistently derived RH.
4. Compare LightGBM against persistence and appropriate alternative regression baselines using identical timestamp alignment and the same chronological test partition.
5. Add automated tests for missing data, wrong units, target-time misalignment, stale input forecasts and failures in the live data source.
6. Keep a model registry and retraining history, monitor drift and define criteria for rejecting a newly trained model if performance deteriorates.
7. Conduct a governed stakeholder review with meteorological experts and evaluate any operational workflow only after appropriate review and approval.
8. Extend the evaluation to other Sri Lankan airports only after suitable local data, features and validation records are available.

These steps would convert the component from a promising prototype with reported summary metrics into a more thoroughly documented and reproducible forecasting service. The highest-priority gaps are direct RH validation, full experiment metadata and performance breakdown by monsoon period.

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GLOSSARY

Dewpoint: The temperature to which air must be cooled, under the relevant conditions, for saturation to occur.

Forecast issue time: The time when a forecast is generated or issued.

Forecast target time: The future time for which the forecast value is intended.

METAR: A coded aerodrome weather observation.

Monsoon: A seasonal circulation pattern associated with a change in prevailing wind and rainfall patterns.

QNH: An aviation pressure setting based on pressure reduced to mean sea level.

Relative humidity: The ratio, expressed as a percentage, between the actual water-vapour pressure and saturation vapour pressure at the same temperature.

Temporal leakage: Use of information from after the forecast issue time in training or evaluation features.

TAF: A coded forecast for aerodrome weather conditions over a specified validity period.

VCBI: The ICAO location indicator for Bandaranaike International Airport.

APPENDICES

Appendix A Research instruments or extended results

This appendix provides a verification checklist for completion against the actual source code and experiment logs. Items below are recommended checks; they are not a claim that a formal test suite was executed.

Table A.1. Component verification checklist

Check ID / activity	Expected outcome or evidence to retain
A-01: Confirm chronological UTC order.	Timestamps are ordered and duplicate records follow the documented rule.
A-02: Check feature generation.	Cyclic time features and rolling statistics match the documented formulas.
A-03: Confirm target shift.	The target is t + 3 hours, not the issue-time value.
A-04: Load the saved model/schema.	Inference feature names, order and units match training.
A-05: Calculate RH for known T and Td.	Output matches the August–Roche–Magnus equation within a stated tolerance.
A-06: Provide missing/stale input.	RH is flagged unavailable/invalid instead of silently displayed as current.
A-07: Persist a forecast.	Issue time, target time and output can be retrieved accurately.
A-08: Verify against later METAR.	Metric calculations use matching variable, units and target timestamps.
A-09: Evaluate by monsoon period.	Per-period scores and sample counts are recorded where data allow.

Reported component results in the supplied research paper: dewpoint MAE ≈ 0.58 °C; dewpoint R^2 ≈ 0.66; QNH MAE ≈ 1.8 hPa. RH-specific accuracy, test sample counts and exact split timestamps must be added only when the relevant output files are available.

Appendix B Academic integrity and AI use disclosure

Tool used: ChatGPT, a generative AI assistant.

Purpose: assistance with organising the thesis in the supplied template, drafting prose from the supplied group research paper, and structuring tables and a component architecture diagram.

Source and verification: numerical results and project-specific technical statements in this draft were taken from the supplied research paper and user-provided details. The thesis writer must verify each statement against project code, experiment logs, primary references and supervisor guidance before submission.

Content affected: initial wording across the thesis chapters, draft requirements and objective tables, the literature summary, and the architecture illustration. The assistant did not generate new experimental observations or claim to rerun the weather models.

Final declaration: this disclosure is a draft and must be updated to match the exact module policy, tools, prompts/workflows and verification process required by SLIIT. The candidate should edit the thesis to reflect their own understanding and must not sign the declaration until the content, references and AI-use disclosure have been reviewed.

Appendix C: Data dictionary and feature catalogue

This appendix defines a reviewable data contract for the individual component. The project paper names the key variables and several feature families, but the exact active schema must be confirmed from the final training and inference code. Suggested fields below are documentation labels; use the repository names in the final submission.

Table C.1. Core observation, forecast and verification fields

Field / feature	Meaning and expected role	Validation / notes
timestamp_utc	Canonical observation time in UTC	Must parse; sort chronologically; inspect duplicates.
qnh_hpa	Observed QNH/pressure target in hPa	Confirm source definition and unit.
dewpoint_c	Observed dewpoint in °C	Retain source timestamp and quality flags.
temperature_c	Observed dry-bulb temperature in °C	Input variable and possible RH reference.
rh_percent	Relative humidity percentage when available	Distinguish direct observation from derived value.
forecast_issue_time	Time inference is generated	Supports input-availability reconstruction.
forecast_valid_time	Time the forecast describes	Should match the declared three-hour horizon.
model_version	Estimator/configuration identifier	Persist with every forecast.
source_name	File/table/API supplying the record	Enables provenance checks.
verification_status	Pending, verified, unavailable or failed	Use a controlled status set.
observed_target	Later observation used for verification	Store source and exact timestamp.
prediction_error	Forecast minus observation, per convention	Calculate only after valid matching.

A minimum record contract separates the forecast issue time from the forecast valid time. This remains necessary even when the expected horizon is three hours because multiple forecasts may target the same valid time, and the project may later test other horizons. Field names in this table should be mapped to the true schema rather than copied into code or treated as confirmed database columns.

Table C.2. Candidate engineered feature families described in the project paper

Feature family	Purpose	Implementation check
Lagged variables	Represent earlier weather observations	Only use timestamps at or before forecast issue time.
Cyclic hour encoding	Represent daily periodicity	Calculate from a consistent timezone.
Cyclic time-of-year encoding	Represent seasonal timing	Document the calendar convention.
Pressure movement	Represent recent QNH trend	State definition, interval and unit.
Three-/six-hour rolling averages	Summarise recent conditions	Exclude future observations from the window.
Dewpoint depression	Temperature minus dewpoint	Ensure time and unit consistency.
Temperature-RH interaction	Nonlinear relation in wider feature set	Confirm whether active in QNH/dewpoint models.

Feature family	Purpose	Implementation check
Wind-speed-RH interaction	Interaction cited for wider system	Do not assume use by this component.
QNH-wind interaction	Combined atmospheric feature	Verify exact active model inputs.
RH squared	Nonlinear humidity transformation	Check availability at inference time.
Present-weather encoding	Categorical weather representation	Avoid using unavailable future codes.
Sequence lookback	Recent sequence used by some system models	Individual LightGBM features may instead be tabular lags.

C.1 Feature manifest

For each active feature, the final artefact should record its output name, input source column, time lag, formula, unit, dtype, missing-value policy, model target and whether the feature is available at live inference. A training manifest should be compared with the inference manifest automatically. This check is particularly important because the group paper describes features for the complete system, while this thesis addresses only QNH, dewpoint and derived RH.

Any feature that depends on a future observation must be excluded from issue-time inputs. Rolling features should be calculated on past records. Scalers or encoders with fitted parameters should be fitted from training data only and versioned with the saved estimator. The final thesis should attach an exported feature list from the actual pipeline when available.

Appendix D: End-to-end procedure and pseudocode

The procedures below formalise the workflow described in the supplied research paper and progress log. They are reference pseudocode for thesis documentation, not an assertion that every listed edge case has already been implemented or tested.

D.1 Historical data preparation

1. Read the approved historical data source and record source name, extraction time, timestamp range and raw row count.
2. Parse timestamps into canonical UTC; quarantine records with invalid timestamps.
3. Sort chronologically and inspect duplicated timestamps. Apply the declared duplicate-resolution rule and log the count.
4. Validate required target variables and units. Mark invalid values for exclusion or correction under a documented policy.
5. Inspect missingness by variable and time period; apply the chosen method without using future values to create earlier features.
6. Construct lagged and time-derived features with the same functions used by the live inference pipeline.
7. Create the three-hour target by aligning input issue time t with the intended target at t+3 hours.
8. Record rows lost to lag construction, missing features and missing targets.
9. Split chronologically without shuffling and retain exact train/test boundary timestamps.
10. Fit any learned transformations on the training partition only, then apply them to later partitions.

D.2 Model training and persistence

11. Load separate configuration for QNH and dewpoint LightGBM estimators.
12. Build X and y using the approved feature manifest and three-hour target definition.
13. Fit each estimator on the training partition; record package versions and key parameters.

14. Predict the chronological holdout and retain issue time and target time beside every prediction.
15. Calculate supported metrics and record the number of valid matched cases.
16. Save the estimator with preprocessing configuration, feature schema, date window, run identifier and code revision.
17. Perform a model-load smoke test in the runtime environment before using the new artefact.

D.3 Live inference and RH calculation

18. Fetch the most recent valid input from the approved source.
19. Check freshness, schema, units, timestamp, missingness and required lag availability.
20. Apply the same versioned preprocessing pipeline as training.
21. Generate QNH and dewpoint forecasts for the defined three-hour target time.
22. Obtain the temperature forecast corresponding to the same target time.
23. Check that temperature and dewpoint are finite numeric values in degrees Celsius and share a valid timestamp.
24. Calculate RH using the configured August-Roche-Magnus relationship and validate the result.
25. Persist an immutable forecast record including issue time, target time, model versions, predictions and RH provenance.
26. Return values with units, target time and status through the API.

D.4 Forecast verification

27. Select forecasts whose target time has passed and whose verification status remains pending.
28. Find the corresponding METAR observation under the defined match rule and tolerance.
29. If no valid observation exists, keep the forecast unverified and record why.
30. If a match exists, save the observation, source, matching timestamp and residual.
31. Ensure repeated verification runs do not create duplicate verification events.
32. Calculate summary metrics over a declared evaluation window and retain sample counts.

33. Review errors by time and monsoon period only where valid data volume supports the analysis.

D.5 Control-flow pseudocode

FOR each forecast issue time t:

observation <- fetch_latest_valid_observation(t)

IF input validation fails: log failure and do not publish a new prediction

features <- shared_preprocessing_pipeline(history available at t)

qnh_hat <- qnh_model.predict(features_qnh)

dewpoint_hat <- dewpoint_model.predict(features_dewpoint)

temperature_hat <- temperature_component prediction for valid_time(t + 3h)

IF temperature_hat is not aligned: mark RH unavailable and log dependency error

ELSE rh_hat <- August_Roche_Magnus(temperature_hat, dewpoint_hat)

save forecast(t, t+3h, values, model versions, input provenance)

END FOR

LATER: match each pending forecast with its target-time METAR observation and update verification.

The pseudocode intentionally uses descriptive operations rather than language-specific code. Any excerpts from the actual repository included in the final thesis should follow university policy and exclude secrets, credentials and unrelated team code.

Appendix E: Evaluation protocol and metric definitions

This appendix defines a repeatable procedure to reproduce the summary scores and complete the missing RH validation. The proposed protocol is not itself evidence that new experiments were run.

E.1 Required evaluation artefacts

Table E.1. Required evaluation artefacts.

Artefact	Required fields	Purpose
Prediction file	Issue/valid time, observed and predicted target, model version	Allows independent metric calculation.
Data split record	Train/test start and end timestamps, row counts	Confirms chronological separation.
Feature manifest	Feature names, units and transformation version	Supports pipeline reproducibility.
Training configuration	Model family, hyperparameters, package version	Documents model fitting.
Evaluation summary	Metric, unit, sample count, evaluation window	Prevents context-free performance claims.
RH provenance	Predicted T, predicted Td, formula/version	Allows RH outputs to be audited.
Verification log	Observation source, match time, status and residual	Supports ongoing evaluation.

E.2 Metric formulae

For n matched cases, let y_i be the observed value and $\hat{y}_i$ the forecast. Mean Absolute Error is $MAE = (1/n) \Sigma |y_i - \hat{y}_i|$. It is expressed in the target unit and represents the mean magnitude of error without regard to direction.

Root Mean Squared Error is $RMSE = \sqrt{(1/n) \Sigma (y_i - \hat{y}_i)^2}$. Since errors are squared before averaging, large errors have more influence than under MAE. It should be reported with the target unit and evaluation sample.

The coefficient of determination is $R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$, where $\bar{y}$ is the mean of the observed values in the evaluated set. An R^2 value is not a percentage of correct predictions; it expresses performance relative to observed variability and the mean reference.

Mean bias can be reported as $MB = (1/n) \Sigma (\hat{y}_i - y_i)$ if the sign convention is explicitly stated. Positive MB then indicates forecasts that tend to be numerically higher than observations, while negative MB indicates lower forecasts. Bias complements MAE because signed errors may cancel.

E.3 Reported component metrics

Table E.2. Reported component metrics.

Target	Metric	Value in paper	Evidence boundary
Dewpoint	MAE	Approximately 0.58 °C	Summary from the paper; exact test count/date range not supplied.
Dewpoint	R^2	Approximately 0.66	Not a “66% accuracy” score.
QNH	MAE	Approximately 1.8 hPa	Reported summary; raw predictions not supplied.
RH	Independent error metric	Not reported	Equation is described/implemented; direct validation remains required.
QNH/dewpoint	RMSE, bias and season-level metrics	Not supplied	Calculate only from actual held-out predictions.

E.4 Metric reproduction procedure

34. Locate the exact saved model versions and prediction outputs associated with the paper.
35. Confirm train/test time boundaries and verify that the evaluated records were not used to fit the model.
36. Join predictions and observations on the same target timestamp using the approved match rule.
37. Apply only predefined exclusion rules and report exclusion counts and reasons.
38. Recompute MAE and R^2 using correct units and compare with the paper’s rounded values.
39. If results differ, check target shifts, row filtering, units, sample definition and rounding before changing the narrative.
40. For RH, use predicted temperature and predicted dewpoint at each eligible holdout timestamp and compare with a compatible reference.
41. Report RH case count, missing inputs, reference definition and MAE/RMSE/bias where available.

E.5 Monsoon-stratified evaluation

A future evaluation should report Southwest Monsoon, Northeast Monsoon, First Inter-Monsoon and Second Inter-Monsoon separately. The exact calendar windows should be approved and stated before calculations. For each period, report matched sample count and appropriate metrics. An aggregate score may hide regime-specific weaknesses, but very small subgroup counts can also produce unstable estimates.

The review should include both ordinary and high-error cases. For each reviewed case, retain the source observation, forecast issue/valid times, predicted values, reference values and missingness flags. Difficult weather records should not be excluded simply because they worsen metrics; exclusions must follow the stated quality policy.

Appendix F: Software and data-quality test matrix

The following matrix is a proposed test plan. The supplied documents do not contain a complete executed software-test report, so status values are left pending. A test should be marked passed only after it is run and its evidence retained.

Table F.1. Software and data-quality test matrix.

ID	Area	Condition	Expected behaviour	Status
DQ-01	Timestamp	Valid UTC timestamp	Parsed to canonical UTC without unintended shift.	To execute
DQ-02	Timestamp	Invalid timestamp string	Rejected or quarantined with reason.	To execute
DQ-03	Duplicates	Rows share a timestamp	Documented duplicate policy is applied and logged.	To execute
DQ-04	Missingness	Required QNH absent	Handled by policy; no silent arbitrary zero.	To execute
DQ-05	Units	Pressure in unexpected scale	Validation flags it before inference.	To execute
DQ-06	Chronology	Rows arrive out of order	Sorted before lag calculation.	To execute
DQ-07	Leakage	Rolling feature at time t	Window uses only data available by t.	To execute
MD-01	Model load	Correct QNH model version	Loads and returns a finite value.	To execute
MD-02	Model load	Model artefact missing	Controlled failure is logged.	To execute
MD-03	Feature contract	Required feature missing	Inference stops with schema error.	To execute
MD-04	Feature contract	Extra feature in request	Explicit schema policy is followed.	To execute
MD-05	Horizon	Input at time t	Target time is correctly aligned to t+3h.	To execute
RH-01	RH formula	Valid same-time T and Td	Finite percentage output is returned.	To execute
RH-02	RH formula	T and Td timestamps differ	RH is flagged unavailable/rejected.	To execute
RH-03	RH formula	Temperature absent	No unsupported RH value is published.	To execute
RH-04	RH formula	Dewpoint is non-numeric	Request fails validation.	To execute
API-01	Response	Valid inference request	Values, units, issue/valid time and status present.	To execute
API-02	Request	Malformed payload	Controlled error; service remains available.	To execute
DB-01	Persistence	Inference succeeds	Forecast record and model version are stored.	To execute
DB-02	Persistence	Database write fails	Service does not claim persistence succeeded.	To execute
VF-01	Verification	Matching METAR available	Correct target observation is linked.	To execute
VF-02	Verification	No matching METAR	Forecast remains unverified, not zero-error.	To execute
VF-03	Verification	Verification job reruns	No duplicate result is created.	To execute
ML-01	MLOps	Candidate training succeeds	Model/data/ configuration metadata are recorded.	To execute
ML-02	MLOps	Retraining fails	Active model remains available.	To execute

ID	Area	Condition	Expected behaviour	Status
ML-03	MLOps	Candidate metrics worsen	Promotion follows documented acceptance rule.	To execute
SEC-01	Security	Credential invalid/missing	Secrets are not exposed in user-facing logs.	To execute
UI-01	Freshness	Forecast becomes stale	UI labels stale state clearly.	To execute

Evidence for a test should include test identifier, date/time, code revision, environment, input fixture, actual output, pass/fail and reviewer. If a test is not relevant to the final implementation, record why. A planned test is useful risk-control documentation but does not prove a test has passed.

Appendix G: Individual contribution and progress-log traceability

This appendix consolidates the dated activities in the supplied Research Project Periodic Log Entry Form for project R26-IT-132. It supports the contribution narrative but does not replace repository history, model artefacts or actual experiment outputs.

Table G.1. Individual contribution and progress-log traceability.

Date	Recorded activity	Thesis relevance / evidence boundary
02 May 2026	Implemented a Random Forest regressor for QNH prediction.	Early model exploration.
03 May 2026	Created an updated dashboard and Pipeline.py.	Pipeline and presentation work.
07 May 2026	Updated backend to support multi-output predictions.	API/output design iteration.
07 May 2026	Added backend API, frontend structure and initial model pipelines.	Initial system integration.
08 May 2026	Added training, verification and feature-importance scripts involving LightGBM and XGBoost.	Model experimentation.
08 May 2026	Improved preprocessing and the XGBoost pipeline.	Data preparation.
09 May 2026	Removed obsolete Random Forest and gradient-boosting pipeline scripts.	Code cleanup; active final path should be verified.
11 May 2026	Prepared PPT1.	Research presentation milestone.
05 June 2026	Reviewed hyper-local forecasting and aviation meteorology literature.	Literature-review work.
10 June 2026	Identified and reviewed additional 2024–2026 weather data.	Data provenance and suitability.
14 June 2026	Analysed monsoon patterns for QNH/dewpoint anomalies.	Seasonal motivation; attach analysis output if available.
22 June 2026	Planned missing-value and data-cleaning strategies.	Method planning; final rules need exact specification.
28 June 2026	Discussed unified verification integration with team.	System integration planning.
08 July 2026	Compared XGBoost, LightGBM and Random Forest theoretically.	Model-selection exploration; no complete numeric comparison supplied.
09 July 2026	Reviewed suitability of the additional dataset.	Data review.
15 July 2026	Designed API architecture and database schema.	Service and storage design.
21 July 2026	Finalised key QNH/dewpoint features.	Feature engineering.
27 July 2026	Met with meteorology department supervisor.	Consultation entry; not itself a formal validation score.
29 July 2026	Drafted MLOps and automated-retraining strategy.	Model maintenance planning.
18 August 2026	Finalised LightGBM three-hour prediction models.	Final model-path milestone recorded in log.
20 August 2026	Created dedicated MLOps auto-retrainer.	Automated model maintenance.
20 August 2026	Added manual inference and live-data verification utilities.	Inference and verification workflow.
20 August 2026	Cleaned obsolete models and updated service configuration.	Release clean-up.
21 August 2026	Moved feature selection into preprocessing pipelines.	Shared transformation contract.
21 August 2026	Implemented RH calculation from predicted dewpoint and temperature.	Formula implementation; target-time alignment still needs validation.
22 August 2026	Removed legacy wind-prediction features from this branch.	Scope control for QNH/dewpoint component.
22 August 2026	Aligned project structure with master branch.	Integration and version alignment.
22 August 2026	Added placeholders for other group members' models.	Cross-component integration readiness.

Date	Recorded activity	Thesis relevance / evidence boundary
25 August 2026	Attended aviation supervisor meeting.	Consultation; detailed feedback is not in the source log.
25 August 2026	Updated individual paper with QNH/dewpoint/RH, database and API details.	Research reporting.
25 August 2026	Completed individual research paper.	Documentation milestone.
28 August 2026	Prepared the second research presentation.	Presentation milestone.

The timeline shows an iterative development process: an early QNH Random Forest prototype, investigation of candidate models, preprocessing and API work, selection of three-hour LightGBM models, RH equation implementation, and MLOps/integration tasks. When describing these activities in the main thesis, planned, implemented, removed and validated tasks must remain distinct. Where possible, the final version should link each major activity to a code revision or saved artefact.

Appendix H: API and data-contract specification for review

The paper identifies FastAPI for backend services, including data retrieval, inference, validation and retraining. Exact route names and payload properties are not given in the supplied paper, so these tables are proposed contract templates. Replace illustrative names with the actual implemented routes and schema before submission.

H.1 Forecast response contract

Table H.1. Forecast response contract.

Property	Type / unit	Expected interpretation
component	String	Identifies QNH/dewpoint/RH component.
issued_at	ISO-8601 UTC	Time inference was generated.
valid_at	ISO-8601 UTC	Time the output describes.
horizon_hours	Integer; expected 3	Declares target horizon.
qnh_prediction	Number; hPa	Three-hour QNH forecast.
dewpoint_prediction	Number; °C	Three-hour dewpoint forecast.
temperature_prediction	Number; °C or null	Temperature value/provenance used for RH.
rh_prediction	Number; percent or null	Only available when inputs are time-aligned.
model_version_qnh	String	QNH estimator identifier.
model_version_dewpoint	String	Dewpoint estimator identifier.
rh_formula_version	String	Equation/code version.
status	Controlled string	Success, partial, unavailable or error as implemented.
verification_status	Controlled string	Pending until a target-time observation is matched.

H.2 Request and response validation

- Reject missing required fields before inference.
- Reject non-finite numerical inputs and invalid timestamp strings.
- Validate availability and ordering of required historical lags.
- Confirm that temperature and dewpoint used in RH calculation share a target timestamp.
- Do not expose stack traces, passwords, API tokens or connection strings in user-facing messages.
- Use a correlation identifier for matching response errors to internal logs.
- Distinguish invalid input, unavailable data, model-loading failure and persistence failure.
- Return explicit partial/unavailable status when RH cannot be calculated.

H.3 Endpoint inventory to reconcile with code

Table H.2. Endpoint inventory to reconcile with code.

Capability	Illustrative route label only	Evidence needed
Current forecast	GET /forecast/qnh-dewpoint	Schema, horizon, units and model versions.
Forecast history	GET /forecasts/history	Time filters and immutable history.
Verification status	GET /verification/status	Pending and verified counts.
Manual inference	POST /inference/run	Input validation and timestamp alignment.
Retraining request	POST /models/retrain	Access control and run status.
Model status	GET /models/status	Active version and last training time.

These route labels are illustrative and must not be represented as actual API routes unless they match the source code. The final thesis should provide the implemented method, path, schema, status codes and access controls based on the version that was tested.

Appendix I: Model card and intended-use statement

This model card summarises scope, reported evidence and limitations for technical review. It does not imply aviation certification or formal operational approval.

Table I.1. Model card and intended-use summary.

Item	Description
Component	QNH, dewpoint and relative humidity prediction model.
Project	Hyper-Local Aviation Weather Forecasting and Verification System for Sri Lankan Monsoon Conditions.
Location scope	Bandaranaike International Airport (VCBI), Sri Lanka.
Forecast horizon	Three hours ahead as described in the supplied paper.
Learned outputs	Separate LightGBM QNH and dewpoint estimates, according to the final paper.
Derived output	RH calculated from predicted dewpoint and temperature using August-Roche-Magnus relationship.
Data scope	Historical VCBI data described as 2019–2026; the overall system paper reports more than 120,000 samples.
Reported results	Dewpoint MAE ≈ 0.58 °C and R^2 ≈ 0.66; QNH MAE ≈ 1.8 hPa.
Not reported	Separate RH score, exact test count/dates, all hyperparameters, season-specific metrics.
Status	Research prototype / decision-support component unless separate approval is documented.

I.1 Intended use

The component is intended for academic research, demonstration and additional airport-local weather context inside the wider system. It can support inspection of forecast history and later verification. Values should be interpreted with their issue time, valid time, units and verification status; this thesis alone does not establish that the outputs are certified operational aviation guidance.

I.2 Out-of-scope use

- Replacing official meteorological observations or forecasts without formal authority and evaluation.
- Guaranteeing flight safety, dispatch suitability or runway decisions from a single model output.
- Generalising VCBI results to another airport without local data and evaluation.
- Claiming RH accuracy before comparison against a compatible holdout reference.
- Claiming stable all-season performance without stratified evaluation and sufficient sample counts.

I.3 Known limitations and failure policy

The supplied sources present summary metrics rather than complete prediction files. Exact feature manifests, configurations and software versions are not included in the research paper. RH has an implemented calculation pathway but no separate reported error metric. Seasonal results, uncertainty estimates, formal stakeholder validation and service-level metrics are also not provided. If input data are stale, a model is unavailable or timestamps are uncertain, the component should report unavailable/partial status and retain a diagnostic record instead of reusing a stale prediction as current.

Appendix J: Reproducibility and final-submission checklist

Use the templates below to complete the final evidence bundle. Empty fields must be filled from actual source files, code or experiment outputs; they must not be guessed.

J.1 Dataset record

Table J.1. Dataset record template.

Field	Value to complete from actual artefact
Training data source/table	[Exact source, table or file]
Extraction time	[Date/time and timezone]
Earliest observation	[YYYY-MM-DD HH:MM UTC]
Latest observation	[YYYY-MM-DD HH:MM UTC]
Raw row count	[Count]
Usable row count	[Count after cleaning]
Duplicate count and policy	[Count and rule]
Missing-value strategy	[Actual rule for each variable]
Train/test boundary	[Exact date/time]
Per-target training/test counts	[Actual counts]
Feature manifest revision	[File path or commit]

J.2 Model record

Table J.2. Model record template.

Field	QNH model	Dewpoint model
Model family	LightGBM; verify in repository	LightGBM; verify in repository
Target	QNH at t+3h	Dewpoint at t+3h
Feature count	[Actual]	[Actual]
Hyperparameters	[Actual configuration]	[Actual configuration]
Training run date	[Actual]	[Actual]
Model artefact/checksum	[Actual]	[Actual]
MAE source	≈1.8 hPa in paper; reproduce	≈0.58 °C in paper; reproduce
R^2 source	Not reported in supplied summary	≈0.66 in paper; reproduce
Holdout count	[Actual]	[Actual]
Code/dependency version	[Commit and lockfile]	[Commit and lockfile]

J.3 RH record

Table J.3. RH reproducibility record.

Field	Value to complete
Formula source path	[Exact function/file]
Formula constants	[Confirm against paper and code]
Temperature forecast source	[Model/version and valid-time]
Time-alignment rule	[Implemented rule]
Reference RH source	[Source/definition]
Eligible RH count	[Count]
RH MAE/RMSE/bias	[Calculate from holdout predictions]
Excluded cases	[Counts and reasons]
Equation unit-test evidence	[Actual test output]

J.4 Final checks

- Recompute every headline metric from retained predictions.
- Confirm consistent reference numbering and complete bibliographic details.
- Insert the final submission month/year and check academic details.

- Remove or fill all placeholders before final submission.
- Update the table of contents, figure/table lists and page numbers after layout is final.
- Ensure every figure and table is cited in the main text.
- Review the AI disclosure against actual module policy and work performed.
- Retain code, data description, configuration, model files and evaluation outputs as required.
- Confirm with the supervisor that appendix material and page count meet the assessment brief.

Appendix K: Monsoon evaluation worksheet and error analysis

This worksheet supports season-aware reporting when actual holdout predictions and matching observations are available. It intentionally contains placeholders instead of invented seasonal scores.

Table K.1. Monsoon evaluation worksheet.

Period	Approved dates	Verified n	QNH MAE (hPa)	Dewpoint MAE (°C)	RH MAE (pp)	Notes
Southwest Monsoon	[Window]	[n]	[value]	[value]	[value]	[Error pattern]
Northeast Monsoon	[Window]	[n]	[value]	[value]	[value]	[Error pattern]
First Inter-Monsoon	[Window]	[n]	[value]	[value]	[value]	[Error pattern]
Second Inter-Monsoon	[Window]	[n]	[value]	[value]	[value]	[Error pattern]
All eligible holdout cases	[Full window]	[n]	[value]	[value]	[value]	Overall result with limits

K.1 Error-case review form

Table K.2. Error-case review form.

Field	Record for each reviewed case
Forecast ID	[Unique forecast ID and model version]
Issue/valid time	[UTC issue time and target time]
Observed source	[METAR or other source and timestamp]
QNH forecast/observation	[Values, hPa error]
Dewpoint forecast/observation	[Values, °C error]
RH forecast/reference	[Values and percentage-point error]
Input completeness	[Missingness, stale record, imputation flags]
Likely issue category	[Model error/data/source/time mismatch/unclear]
Corrective action	[Action, owner and date]

Review ordinary and high-error examples under a declared sampling policy. Avoid selecting only cases that support the model. If any cases are excluded from a metric, state the rule and count. Seasonal comparisons require approved season definitions and a consistent verification match policy.

Appendix L: Operational risk register and runbook

These entries convert the documented dependencies into practical controls. They are recommended controls for a research component and do not assert that every control is already deployed.

Table L.1. Operational risk register.

Risk	Consequence	Detection	Recommended control
Stale weather input	Forecast uses outdated state.	Track source time and age.	Define freshness threshold; label or block stale inference.
Target-time error	Wrong observation is scored.	Audit issue/valid-time differences.	Canonical UTC and horizon tests.
Feature mismatch	Inference differs from training.	Compare schema and version.	Shared versioned preprocessing.
Bad imputation	Artificial patterns or leakage.	Log imputation counts.	Per-variable documented rule; avoid future leakage.
Model drift	Forecast quality declines.	Monitor rolling verification metrics.	Review and retrain with acceptance checks.
Failed retraining	New model unavailable/faulty.	Capture run status.	Keep current model until candidate passes.
RH misalignment	Inconsistent derived output.	Compare input valid times.	Reject or mark RH unavailable.
Missing verification	Metrics incomplete or biased.	Count pending cases.	Retain unverified status and report coverage.
Database failure	Forecast is not retained.	Check persisted record.	Separate inference success from persistence success.
Credential leak	Unauthorised access.	Inspect logs and repository.	Use protected secrets and redacted errors.
Untraceable model version	Old results cannot be reconstructed.	Inspect forecast metadata.	Persist model and preprocessing identifiers.
Overstated operational use	Prototype mistaken as certified.	Review UI labels.	State decision-support status and limitations.

L.1 Suggested operational check

42. Confirm that the latest valid METAR has arrived and meets the configured freshness policy.
43. Check the inference job and confirm the expected forecast record exists.
44. Inspect QNH/dewpoint outputs, units and the declared target horizon.
45. Confirm the RH temperature and dewpoint are aligned to the same target time.
46. Check API/database health separately from prediction-quality metrics.
47. Review forecasts that passed their valid time but remain unverified.
48. Check rolling metrics only when the verified sample count is adequate.
49. Escalate repeated faults; do not hide them by labelling stale forecasts as current.

L.2 Monthly retraining review

50. Record training data window and confirm VCBI coverage.
51. Run data-quality and timestamp checks before fitting.
52. Train candidate models using the versioned feature pipeline.

53. Evaluate against an independent chronological validation period.
54. Compare candidate and current-model scores on identical cases.
55. Confirm model artefacts load and response schema remains compatible.
56. Promote only if the stated acceptance policy is satisfied.
57. Save data window, configuration, metrics, code version and review decision.

Appendix M: RH equation notes and interpretation boundaries

This appendix explains the dependency between RH and the forecast inputs. The exact equation and implementation should be checked against the actual code and the supplied research paper before submission.

M.1 Temperature and dewpoint relationship

Let T denote dry-bulb temperature in degrees Celsius and Td denote dewpoint temperature in degrees Celsius. The form stated in the supplied paper is:

$$RH = 100 \times \exp[(17.67 \times Td / (Td + 243.5)) - (17.67 \times T / (T + 243.5))].$$

The equation returns RH as a percentage when the inputs and constants are used consistently. Because RH is derived from predictions, its error depends on temperature and dewpoint forecast errors and the implementation. A correct formula can still produce an inaccurate future RH if the upstream forecasts are inaccurate or not time-aligned.

M.2 Implementation checks

- Confirm T and Td are in degrees Celsius, not Kelvin.
- Confirm both input values describe the same target valid time.
- Handle missing, non-finite and non-numeric inputs explicitly.
- Retain the formula version or code revision with generated RH records.
- Compare the function against independently checked sample values.
- Calculate RH metrics against a compatible holdout reference before claiming accuracy.
- Document any policy for flagging or bounding abnormal outputs.

M.3 Interpreting derived values

The calculated RH is a dependent output; it is not independent evidence that temperature and dewpoint forecasts are correct. It is not equivalent to a measured RH unless calculated from suitable observed inputs. Validation must compare forecast-derived RH to a direct or consistently derived reference at the same valid time and must state which reference was used.

Appendix N: Final result-reporting templates

Complete these tables only with numbers recomputed from saved predictions or directly supported by a reviewed experiment report. Placeholder cells should be filled or removed before final submission.

N.1 Overall result table

Table N.1. Overall result-reporting template.

Target	Test window	n	MAE	RMSE	R^2	Bias	Run/source ID
QNH	[insert]	[n]	[hPa]	[hPa]	[value]	[hPa]	[run ID]
Dewpoint	[insert]	[n]	[°C]	[°C]	[value]	[°C]	[run ID]
Derived RH	[insert]	[n]	[pp]	[pp]	[value if appropriate]	[pp]	[run ID]

N.2 Baseline comparison table

Table N.2. Baseline comparison template.

Target	Method	Same test n	MAE	RMSE	Evaluation note
QNH	Persistence baseline	[n]	[hPa]	[hPa]	Same target-time cases.
QNH	Final LightGBM	[n]	[hPa]	[hPa]	Use actual prediction file.
Dewpoint	Persistence baseline	[n]	[°C]	[°C]	Same target-time cases.
Dewpoint	Final LightGBM	[n]	[°C]	[°C]	Use actual prediction file.
RH	Derived from predicted T/Td	[n]	[pp]	[pp]	Use compatible RH reference.

These templates identify what is needed for reproducible reporting; they are not permission to guess missing scores. If the files are unavailable, report the limitation and rerun evaluation where feasible.